



Solitons, breathers and rational solutions for a (2+1)-dimensional dispersive long wave system

Han-Han Sheng, Guo-Fu Yu^{*}

School of Mathematical Sciences, CMA-Shanghai, Shanghai Jiao Tong University, Shanghai 200240, PR China

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ABSTRACT

In this paper, we investigate a (2+1)-dimensional dispersive long wave system with an arbitrary constant a . Solitons, line breathers, rational solutions and algebraic solitons of the system are constructed based on Hirota's bilinear method and Kadomtsev–Petviashvili (KP) hierarchy reduction technique. These solutions are expressed in terms of $N \times N$ determinant. When the size of the determinant N is odd, solutions on periodic backgrounds are generated. Solutions on constant backgrounds are derived when N is even. By using asymptotic analysis, we elucidate explicit expressions of asymptotic algebraic solitons localized in the straight for the algebraic soliton solutions. As a byproduct, we find lump solutions for an intermediate potential of the system. The effects of constant a are discussed and it is shown that the constant can affect collision of solitons. Dynamics of the obtained solutions are discussed with plots.

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1. Introduction

In the 1960s, Zabusky and Kruskal used finite difference approach to study numerical solutions of KdV equation and introduced the concept of solitons [1]. Solitons have the prototype elastic collision property, that is, the shapes and velocities keep the same after collisions. Besides KdV equation, solitons exist in other (1+1)-dimensional continuous and discrete equations, including the modified KdV, the Burgers equation, the sine–Gordon equation, the nonlinear Schrödinger equation and their higher dimensional extensions. These equations have been widely used to model nonlinear phenomena in physics, optics, biology and chemistry. Studies on solitons are very active and the topic has attracted great attention. With the development of soliton theory, several types of soliton solutions are found, such as simple line solitons, loop solitons, algebraic solitons, breathers, rogue waves, cuspon and peakon solutions, et al. For (2 + 1)-dimensional integrable equations, for example, Davey–Stewartson (DS) equation and KP equation, there exist localized soliton solutions, including dromions, lumps and rogue wave solutions. Although mathematical structures and hidden algebraic properties of soliton equations have attracted more and more attention, investigation of solutions of nonlinear soliton equations is a permanent topic because these solutions can help us understand nonlinear phenomena.

The structure of soliton equations has been clarified due to various analytic and algebraic methods, such as inverse scattering method [2], Riemann–Hilbert method [3], Darboux transformation method [4] and Hirota bilinear method [5]. Among those methods, Hirota bilinear method is purely algebraic and direct. By transforming soliton equations into bilinear forms, we can not only derive N -soliton solutions and N -periodic wave solutions, but also obtain integrable characteristics such as Bäcklund transformation, Lax pairs and infinite conservation laws. In general, these soliton solutions own determinant or Pfaffian structures. Sato theory gives deep understanding of Hirota's method and present bilinear integrable hierarchy [6]. Recently, breathers and rogue waves have attracted much attention. There are two kinds of breathers, spacial periodic and time periodic, respectively. In the limit case, breathers could become rogue waves. Rogue waves originate from oceanography [7] and describe an instantaneous excitation phenomenon. Rogue wave solutions of integrable systems are rational in both space and time variables. Based on Lax pairs, Darboux transformation method is useful to construct rational solutions [8–10]. Besides, the KP hierarchy reduction technique based on Hirota bilinear method is also a powerful approach to generate rational solutions of integrable systems [11–14]. This method has been successfully applied to study solutions of nonlocal integrable systems [15,16]. Algebraic solitons are some kind of rational solutions and less studied for soliton

^{*} Corresponding author.

E-mail address: gfyu@sjtu.edu.cn (G.-F. Yu).

equations. They appear in KP equation [17], DS equation [18], the Benjamin–Ono equation [19], a model equation for deep water waves [20], and also in some nonlocal integrable systems [21,22]. Algebraic solitons also exhibit elastic collision phenomena. It is interesting to find algebraic soliton solutions for soliton equations. Meanwhile, solutions of integrable systems are mainly on constant backgrounds, zero or nonzero, and less discussion about solutions on the periodic backgrounds.

The aim of the present paper is devoted to the studies of a (2+1)-dimensional dispersive long wave (DLW) system

$$u_{yt} + v_{xx} + uu_{xy} + u_x u_y + 2au_{xy} = 0, \quad (1)$$

$$v_t + (uv)_x + u_{xy} + 2av_x + 2\lambda u_x = 0, \quad (2)$$

where λ and a are arbitrary constants. By the variable transformation $x \rightarrow x - 2at$, DLW system (1)–(2) reduces to

$$u_{yt} + v_{xx} + uu_{xy} + u_x u_y = 0, \quad (3)$$

$$v_t + (uv)_x + u_{xy} + 2\lambda u_x = 0, \quad (4)$$

which was firstly derived by Boiti in 1987 [23] to model the hydrodynamics of a wide channel or open sea with a finite depth. Here $u(x, y, t)$ is the horizontal velocity and $v(x, y, t)$ is the wave height above the undisturbed water surface. Variables t and (x, y) represent the time and propagation plane, respectively. Under the dimension reduction $y = x$, one dimensional DLW system can approximate the inviscid gravity waves on a water layer with constant depth [24,25].

DLW system has attracted great attention. In [26], Grammian and Wronskian type determinant solutions of DLW system were presented based on the KP reduction approach. By choosing exponential type determinant elements, N -bright soliton solutions of DLW were derived [26]. Group theoretical analysis [27], non-Painlevé property [28], Darboux transformation [29,30], Kac–Moody–Virasoro symmetry algebra [31] and line solitons and dromion solutions [32–34] of DLW system were studied. We will present different kinds of solutions of DLW system and show that soliton solutions for DLW system given in [34] is a constrained case of our soliton solutions derived here. Recently, Grinevich and Santini [35] show that the rogue wave solution or the Akhmediev breather solution on a nonzero background expressed by elliptic function can be formed. Breather and rogue wave solutions for the Hirota equation and nonlinear Schrödinger equation on the elliptic function background were studied [36,37]. The construction of soliton solutions on periodic function backgrounds is much more challenge. In this paper, one aim is to construct general solutions, such as solitons, breathers and algebraic soliton solutions, of DLW system on the periodic function background systematically. Compared with asymptotic analysis of exponential-type solitons, asymptotic behavior of algebraic soliton solutions is much more difficult to be conducted. We analyze the asymptotic behaviors of algebraic solitons of DLW system and show that wave peaks evolve along the lines.

The paper is organized as follows. In Section 2, we introduce the derivation and reduction of the solutions of DLW system from the modified KP equations. In Section 3, we analyze the dynamics of solitons on constant and periodic background, respectively. In Section 4, we investigate the breather solutions. In Section 5, we discuss the dynamic behaviors of the rational solutions which include algebraic solitons, multiple and high-order rational solutions. Lump solutions are found for some intermediate potential. Section 6 is devoted to the conclusion and discussion.

2. Soliton and (semi-)rational solutions in determinant form

The spirit of Hirota bilinear method is to transform nonlinear equations into bilinear form by dependent variable transformation. Firstly, we make the dependent variable transformation

$$u = 2 \left(\ln \frac{f}{g} \right)_x, \quad v = 2(\ln fg)_{xy}, \quad (5)$$

and transform DLW system (1)–(2) into the bilinear equations

$$[D_y(D_x^2 + D_t + 2aD_x) + 2\lambda D_x]f \cdot g = 0, \quad (6)$$

$$(D_x^2 + D_t + 2aD_x)f \cdot g = 0. \quad (7)$$

The Hirota's bilinear operator D is defined by

$$D_x^n f(x) \cdot g(x) = \frac{\partial^n}{\partial \delta^n} f(x + \delta) g(x - \delta) \Big|_{\delta=0}. \quad (8)$$

We note that bilinear DLW system is very similar to bilinear (2 + 1)-dimensional sinh-Gordon equation. In fact, under the variable interchange $y \leftrightarrow t$ and $a = 0$, $\lambda = 1$, the bilinear DLW system becomes bilinear (2 + 1)-dimensional sinh-Gordon equation [38,39]. It is common that different nonlinear integrable equations can share same bilinear equations by different dependent variable transformation. For example, the bilinear form of the (2 + 1)-dimensional Gardner equation is the same as the bilinear modified KP equation [40]. In [26], bright soliton solutions with the determinant structure were derived for (2 + 1)-dimensional sinh-Gordon equation. Rational solutions on the zero background with the determinant structure for DLW system were presented in [39].

In the following discussion we take the parameter $\lambda = -2$ for convenient. We start from the bilinear modified KP equations

$$(D_{x_1}(D_{x_1}^2 - D_{x_2} + 2aD_{x_1}) - 4D_{x_1})\tau_{n+1} \cdot \tau_n = 0, \quad (9)$$

$$(D_{x_1}^2 - D_{x_2} + 2aD_{x_1})\tau_{n+1} \cdot \tau_n = 0. \quad (10)$$

Lemma 2.1 ([41]). *The modified KP equations (9)–(10) have the Gram-type determinant solution*

$$\tau_n = \det_{1 \leq i, j \leq N} (m_{ij}^{(n)}), \quad (11)$$

where the determinant elements satisfy the relations

$$\begin{aligned}\partial_{x_1} m_{ij}^{(n)} &= \phi_i^{(n)} \cdot \psi_j^{(n)} \\ \partial_{x_2} m_{ij}^{(n)} &= \phi_i^{(n+1)} \cdot \psi_j^{(n)} + \phi_i^{(n)} \cdot \psi_j^{(n-1)} + 2a\phi_i^{(n)} \cdot \psi_j^{(n)} \\ \partial_{x_{-1}} m_{ij}^{(n)} &= -\phi_i^{(n-1)} \cdot \psi_j^{(n+1)} \\ m_{ij}^{(n+1)} &= m_{ij}^{(n)} + \phi_i^{(n)} \cdot \psi_j^{(n+1)} \\ \partial_{x_2} \phi_i^{(n)} &= \phi_i^{(n+2)} + 2a\phi_i^{(n+1)} + a^2\phi_i^{(n)} \\ \partial_{x_1} \phi_i^{(n)} &= \phi_i^{(n+1)} + a\phi_i^{(n)} \\ \partial_{x_{-1}} \phi_i^{(n)} &= \phi_i^{(n-1)} \\ \partial_{x_2} \psi_j^{(n)} &= -\psi_j^{(n-2)} - 2a\psi_j^{(n-1)} - a^2\psi_j^{(n)} \\ \partial_{x_1} \psi_j^{(n)} &= -\psi_j^{(n-1)} - a\psi_j^{(n)} \\ \partial_{x_{-1}} \psi_j^{(n)} &= -\psi_j^{(n+1)}.\end{aligned}$$

Here $m_{ij}^{(n)}$, $\phi_i^{(n)}$, $\psi_j^{(n)}$ are functions of x_{-1} , x_1 , x_2 and N is any positive integer.

In general, we choose the following special ϕ and ψ functions to construct soliton solutions,

$$\phi_i^{(n)} = (p_i - a)^n e^{\xi_i}, \quad \psi_j^{(n)} = (-q_j - a)^{-n} e^{\eta_j}, \quad (12)$$

$$m_{ij}^{(n)} = c_j \delta_{i,j} + \frac{1}{p_i + q_j} \left(-\frac{p_i - a}{q_j + a} \right)^n e^{\xi_i + \eta_j}, \quad (13)$$

$$\xi_i = p_i x_1 + p_i^2 x_2 + \frac{1}{p_i - a} x_{-1} + \xi_{i0}, \quad (14)$$

$$\eta_j = q_j x_1 - q_j^2 x_2 + \frac{1}{q_j + a} x_{-1} + \eta_{j0}, \quad (15)$$

where c_j , p_i , q_j and ξ_{i0} , η_{j0} are arbitrary parameters. Under the variable transformation $x_{-1} = y$, $x_1 = x$, $x_2 = -t$ and $f = \tau_{n+1}$, $g = \tau_n$, the modified KP equation (9)–(10) turns into bilinear DLW system (6)–(7). Substituting above elements $m_{ij}^{(n)}$ into the Gram determinant, we obtain N -soliton solutions for DLW system.

Remark 2.1. One can see that the definition of elements $m_{ij}^{(n)}$ here is very similar to that of (2+1)-dimensional sinh-Gordon equation in [39] and the differences mainly lie in the singular shift $p_j - a$ and $q_j + a$ besides the variable interchange $y \leftrightarrow t$.

Following the method proposed in [11], we introduce the differential operators A_i and B_j

$$A_i = \sum_{k=0}^{n_i} a_{ik} \left((p_i - a) \partial_{p_i} \right)^{n_i - k}, \quad B_j = \sum_{l=0}^{n_j} b_{jl} \left((q_j + a) \partial_{q_j} \right)^{n_j - l}, \quad (16)$$

to generate rational solutions. Here a_{ik} and b_{jl} are constants. We apply the differential operators A_i and B_j to $m_{ij}^{(n)}$ and denote

$$\begin{aligned}\tilde{m}_{ij}^{(n)} &= A_i B_j m_{ij}^{(n)} \\ &= c_j \delta_{i,j} + \left(-\frac{p_i - a}{q_j + a} \right)^n e^{\xi_i + \eta_j} \sum_{k=0}^{n_i} a_{ik} \left((p_i - a) \partial_{p_i} + n + \xi_i \right)^{n_i - k} \\ &\quad \times \sum_{l=0}^{n_j} b_{jl} \left((q_j + a) \partial_{q_j} - n + \eta_j \right)^{n_j - l} \frac{1}{p_i + q_j},\end{aligned} \quad (17)$$

with

$$\tilde{\xi}_i = (p_i - a)x_1 + 2p_i(p_i - a)x_2 - \frac{1}{p_i - a}x_{-1}, \quad (18)$$

$$\tilde{\eta}_j = (q_j + a)x_1 - 2q_j(q_j + a)x_2 - \frac{1}{q_j + a}x_{-1}. \quad (19)$$

Because these operators apply only to parameters p_i and q_j and thus the determinant $\tau_n = \det(\tilde{m}_{ij}^{(n)})_{1 \leq i, j \leq N}$ also solves bilinear equations (9)–(10). Furthermore, according to the gauge freedom of τ_n , we introduce $\tilde{p}_i = p_i - a$, $\tilde{q}_j = q_j + a$ and rewrite the Gram determinants $\tau_n = \det(\tilde{m}_{ij}^{(n)})$ of (9)–(10) with determinant elements

$$\tilde{m}_{ij}^{(n)} = c_j \delta_{i,j} e^{-\tilde{\xi}_i - \tilde{\eta}_j} + \left(-\frac{\tilde{p}_i}{\tilde{q}_j} \right)^n \sum_{k=0}^{n_i} a_{ik} (\tilde{p}_i \partial_{\tilde{p}_i} + n + \tilde{\xi}_i)^{n_i - k} \times \sum_{l=0}^{n_j} b_{jl} (\tilde{q}_j \partial_{\tilde{q}_j} - n + \tilde{\eta}_j)^{n_j - l} \frac{1}{\tilde{p}_i + \tilde{q}_j}. \quad (20)$$

When all c_j equal zero in (20), determinant elements are all polynomials of x , y , t that lead τ_n to be polynomials, and we obtain rational solutions of DLW system. While if $c_j \neq 0$, we can get semi-rational solutions expressed by the combination of exponential functions and polynomials. To summarize, we have the following theorem.

Theorem 2.1. DLW system (1)–(2) admits solution

$$u = 2 \left(\ln \frac{\tau_{n+1}}{\tau_n} \right)_x, \quad v = 2 (\ln \tau_n \tau_{n+1})_{xy}, \quad (21)$$

where the determinant $\tau_n = \det_{1 \leq i, j \leq N} (\tilde{m}_{ij}^{(n)})$ has elements

$$\tilde{m}_{ij}^{(n)} = c_j \delta_{i,j} e^{-\xi_i - \eta_j} + \left(-\frac{\tilde{p}_i}{\tilde{q}_j} \right)^n \sum_{k=0}^{n_i} c_{ik} (\tilde{p}_i \partial_{\tilde{p}_i} + n + \tilde{\xi}_i)^{n_i-k} \times \sum_{l=0}^{n_j} d_{jl} (\tilde{q}_j \partial_{\tilde{q}_j} - n + \tilde{\eta}_j)^{n_j-l} \frac{1}{\tilde{p}_i + \tilde{q}_j} \quad (22)$$

with

$$\tilde{\xi}_i = \tilde{p}_i x - 2\tilde{p}_i(\tilde{p}_i + a)t - \frac{1}{\tilde{p}_i} y \quad (23)$$

$$\tilde{\eta}_j = \tilde{q}_j x + 2\tilde{q}_j(\tilde{q}_j - a)t - \frac{1}{\tilde{q}_j} y \quad (24)$$

$$\xi_i = (\tilde{p}_i + a)x - (\tilde{p}_i + a)^2 t + \frac{1}{\tilde{p}_i} y + \xi_{i0} \quad (25)$$

$$\eta_j = (\tilde{q}_j - a)x + (\tilde{q}_j - a)^2 t + \frac{1}{\tilde{q}_j} y + \eta_{j0}. \quad (26)$$

Remark 2.2. Under the variable transformation $y \rightarrow -\frac{1}{2}y$ and the interchange $y \leftrightarrow t$, when $a = 0$ and $c_j = 0$ ($j = 1, 2, \dots, N$) the tau function solution τ_n with elements (22)–(26) degenerates to solutions of the (2+1)-dimensional sinh-Gordon equation given in [39].

3. Dynamics of the soliton solutions

In this part, we study the dynamics of soliton solutions u and v of DLW system with constant and periodic background, respectively. First, we elucidate the difference between soliton solutions for DLW system here and soliton solutions for DLW system obtained in [26,34].

- (1) Setting $a = 0$, $c_j = 1$, the determinant solution of DLW with elements (13)–(15) is reduced to soliton solutions in [26].
- (2) Setting $a = 0$, $c_j = 1$, variables transformation $p_i = -\hat{p}_i$, $q_j = \hat{q}_j$, and the constraint $\exp(\xi_{i0} + \eta_{j0}) = (\hat{p}_i + \hat{q}_j)\hat{q}_j$, the determinant solution of DLW with elements (13)–(15) is degenerated into soliton solutions in [34] with variables $\hat{p}$, $\hat{q}$.

3.1. One-soliton solution on the zero and periodic backgrounds

We set $N = 1$ in (13)–(15) and get first order determinant

$$m_{11}^{(n)} = c_1 + \left(-\frac{p_1 - a}{q_1 + a} \right)^n \frac{1}{p_1 + q_1} e^{\xi_1 + \eta_1}, \quad n = 0, 1,$$

$$\xi_1 + \eta_1 = (p_1 + q_1)x + (q_1^2 - p_1^2)t + \left(\frac{1}{q_1 + a} + \frac{1}{p_1 - a} \right)y + \xi_{10} + \eta_{10}.$$

Here $p_1, q_1, c_1, \xi_{10}, \eta_{10}$ are arbitrary constants. We take $c_1 = 1$ and the potential u is of the form

$$u = 2 \left(\ln \frac{f}{g} \right)_x = -\frac{2(p_1 + q_1)^2}{2\sqrt{(a - p_1)(a + q_1)} \cosh(\xi_1 + \eta_1 - \theta) + (2a - p_1 + q_1)}, \quad (27)$$

where $\exp(\theta) = (p_1 + q_1)\sqrt{\frac{a+q_1}{a-p_1}}$. Here we require that both $p_1 + q_1$ and $(a - p_1)(a + q_1)$ to be positive to avoid the singularity. Plots of one-bright-soliton solution (27) are shown in Fig. 1.

When we take $c_1 = ci$, $p_1 = \alpha + \beta i$ and $q_1 = -p_1^*$, functions f and g have the form

$$f = c_1 + \left(-\frac{p_1 - a}{q_1 + a} \right) \frac{1}{p_1 + q_1} e^{\xi_1 + \eta_1} = ci - \frac{\alpha - a + \beta i}{2\beta i(-\alpha + a + \beta i)} \exp \left[2\beta i \left(x - 2\alpha t - \frac{y}{(\alpha - a)^2 + \beta^2} \right) + \xi_{10} + \eta_{10} \right],$$

$$g = c_1 + \frac{1}{p_1 + q_1} e^{\xi_1 + \eta_1} = ci - \frac{i}{2\beta} \exp \left[2\beta i \left(x - 2\alpha t - \frac{y}{(\alpha - a)^2 + \beta^2} \right) + \xi_{10} + \eta_{10} \right].$$

Actually, f and g are periodic in this case and both $u = 2 \left(\ln \frac{f}{g} \right)_x$ and $v = 2(\ln fg)_{xy}$ are periodic solution. Furthermore, under the special circumstance $\alpha = a$, we obtain one-soliton solution

$$u = 2 \left(\ln \frac{f}{g} \right)_x = \frac{16i\beta^2 c \exp(2i\zeta - \xi_{10} - \eta_{10})}{4\beta^2 c^2 \exp(4i\zeta - 2\xi_{10} - 2\eta_{10}) - 1}, \quad (28)$$

where $\zeta = 2a\beta t - \beta x + \beta^{-1}y$. It is clear that u is regular when $|2\beta c \exp(-\xi_{10} - \eta_{10})| \neq 1$. In this case, the solution (28) is periodic with period π/β in x , $\pi\beta$ in y and $\pi/(a\beta)$ in t . Plot of this periodic one-soliton solution is depicted in Fig. 2.

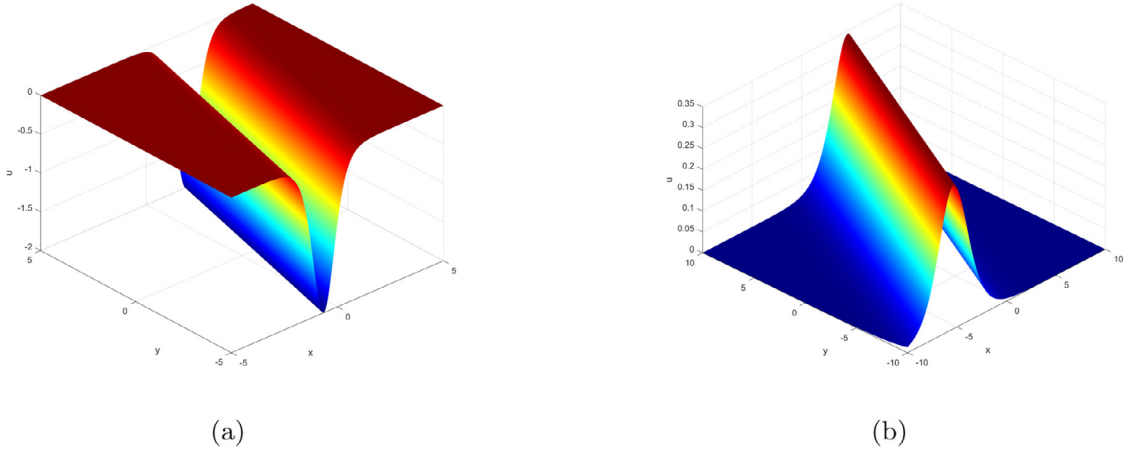


Fig. 1. One-soliton solution u with parameters (a) $p_1 = 1, q_1 = 2, t = 1, a = 2, \xi_1 = 0, \eta_1 = 0, c_1 = 1$, and (b) $p_1 = 4, q_1 = -3, t = 0, a = 2, \xi_1 = 0, \eta_1 = 0, c_1 = 1$.

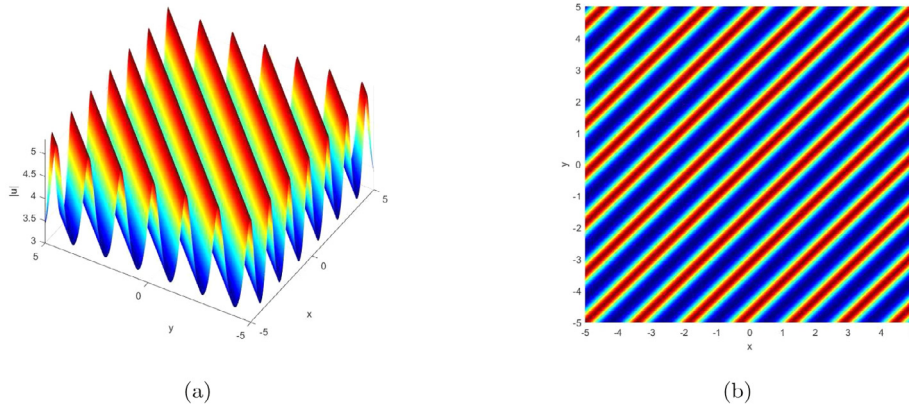


Fig. 2. The periodic solution u (28) with parameters $\beta = 1, c = 1, \xi_{10} = \eta_{10} = 0, a = \alpha = 1, t = 0$. (b) is the corresponding density plot of u .

3.2. Two-soliton solutions on the zero background

To derive two-soliton solution, we set $N = 2$ in (11). For convenience, we take $c_1 = c_2 = 1$. Then we have the tau function solutions f and g in 2×2 determinant

$$f = \begin{vmatrix} 1 + \frac{1}{p_1+q_1} \left(-\frac{p_1-a}{q_1+a} \right) e^{\xi_1+\eta_1} & \frac{1}{p_1+q_2} \left(-\frac{p_1-a}{q_2+a} \right) e^{\xi_1+\eta_2} \\ \frac{1}{p_2+q_1} \left(-\frac{p_2-a}{q_1+a} \right) e^{\xi_2+\eta_1} & 1 + \frac{1}{p_2+q_2} \left(-\frac{p_2-a}{q_2+a} \right) e^{\xi_2+\eta_2} \end{vmatrix} \quad (29)$$

$$= 1 - \frac{1}{p_1+q_1} \frac{p_1-a}{q_1+a} e^{\xi_1} - \frac{1}{p_2+q_2} \frac{p_2-a}{q_2+a} e^{\xi_2} + \frac{(p_1-p_2)(q_1-q_2)}{(p_1+q_1)(p_1+q_2)(p_2+q_1)(p_2+q_2)} \frac{(p_1-a)(p_2-a)}{(q_1+a)(q_2+a)} e^{\xi_1+\xi_2},$$

$$g = \begin{vmatrix} 1 + \frac{1}{p_1+q_1} e^{\xi_1+\eta_1} & \frac{1}{p_1+q_2} e^{\xi_1+\eta_2} \\ \frac{1}{p_2+q_1} e^{\xi_2+\eta_1} & 1 + \frac{1}{p_2+q_2} e^{\xi_2+\eta_2} \end{vmatrix} \quad (30)$$

$$= 1 + \frac{1}{p_1+q_1} e^{\xi_1} + \frac{1}{p_2+q_2} e^{\xi_2} + \frac{(p_1-p_2)(q_1-q_2)}{(p_1+q_1)(p_1+q_2)(p_2+q_1)(p_2+q_2)} e^{\xi_1+\xi_2},$$

where

$$\zeta_i = \xi_i + \eta_i = (p_i + q_i)x + (q_i^2 - p_i^2)t + \left(\frac{1}{q_i + a} + \frac{1}{p_i - a} \right) y + \xi_{i0} + \eta_{i0}, \quad i = 1, 2. \quad (31)$$

We analyze the collision of two solitons and give asymptotic behaviors. For convenience, we suppose $t = 0$ in the following discussion. Two-soliton solution of u is expressed as

$$u = 2 \left(\ln \frac{f}{g} \right)_x = 2 \frac{A_{112} e^{2\xi_1+\xi_2} + A_{122} e^{\xi_1+2\xi_2} + A_{12} e^{\xi_1+\xi_2} + A_1 e^{\xi_1} + A_2 e^{\xi_2}}{B_{1122} e^{2\xi_1+2\xi_2} + B_{112} e^{2\xi_1+\xi_2} + B_{122} e^{\xi_1+2\xi_2} + B_{12} e^{\xi_1+\xi_2} + B_{11} e^{\xi_1} + B_{22} e^{2\xi_2} + B_1 e^{\xi_1} + B_2 e^{\xi_2} + 1},$$

$$A_1 = -\frac{p_1+q_1}{q_1+a}, \quad A_2 = -\frac{p_2+q_2}{q_2+a},$$

$$A_{112} = -\frac{(p_2+q_2)(-p_1+a)(p_1-p_2)(q_1-q_2)}{(p_1+q_1)^2(p_1+q_2)(p_2+q_1)(q_1+a)(q_2+a)},$$

$$\begin{aligned}
A_{122} &= -\frac{(p_1 + q_1)(-p_2 + a)(p_1 - p_2)(q_1 - q_2)}{(p_2 + q_2)^2(p_1 + q_2)(p_2 + q_1)(q_1 + a)(q_2 + a)}, \\
B_1 &= \frac{-p_1 + q_1 + 2a}{(p_1 + q_1)(q_1 + a)}, \quad B_2 = \frac{-p_2 + q_2 + 2a}{(p_2 + q_2)(q_2 + a)}, \\
B_{11} &= \frac{-p_1 + a}{(p_1 + q_1)^2(q_1 + a)}, \quad B_{22} = \frac{-p_2 + a}{(p_2 + q_2)^2(q_2 + a)}, \\
B_{112} &= \frac{(-p_1 + a)(p_1 - p_2)(q_1 - q_2)(-p_2 + 2a + q_2)}{(p_2 + q_1)(p_1 + q_2)(q_1 + a)(q_2 + a)(p_1 + q_1)^2(p_2 + q_2)}, \\
B_{122} &= \frac{(-p_2 + a)(p_1 - p_2)(q_1 - q_2)(-p_1 + 2a + q_1)}{(p_2 + q_1)(p_1 + q_2)(q_1 + a)(q_2 + a)(p_1 + q_1)(p_2 + q_2)^2}, \\
B_{1122} &= \frac{(-p_1 + a)(-p_2 + a)(p_1 - p_2)^2(q_1 - q_2)^2}{(p_2 + q_1)^2(p_1 + q_2)^2(q_1 + a)(q_2 + a)(p_1 + q_1)^2(p_2 + q_2)^2}.
\end{aligned}$$

Without loss of generality, we assume that $p_i + q_i > 0$, ($i = 1, 2$) and $(-p_2 + a)(q_2 + a) > (-p_1 + a)(q_1 + a) > 0$. We denote the left-moving soliton along the plane $\zeta_1 = 0$ by soliton 1 and the right-moving soliton along the plane $\zeta_2 = 0$ by soliton 2. By conducting the asymptotic analysis, we have the following asymptotic forms for the two-bright-soliton solution.

(1) Before collision ($y \rightarrow -\infty$).

Soliton 1 ($\zeta_1 \approx 0, \zeta_2 \rightarrow +\infty$):

$$\begin{aligned}
u &\rightarrow \frac{2A_{122}e^{\zeta_1}}{B_{22} + B_{122}e^{\zeta_1} + B_{1122}e^{2\zeta_1}} = \frac{-2(p_1 + q_1)^2}{(-p_1 + q_1 + 2a) + 2\sqrt{(-p_1 + a)(q_1 + a)}\cosh(\zeta_1 - \theta_1)}, \\
\exp(\theta_1) &= \frac{(p_2 + q_1)(p_1 + q_2)(p_1 + q_1)}{(p_1 - p_2)(q_1 - q_2)} \sqrt{\frac{q_1 + a}{-p_1 + a}}.
\end{aligned}$$

Soliton 2 ($\zeta_2 \approx 0, \zeta_1 \rightarrow -\infty$):

$$\begin{aligned}
u &\rightarrow \frac{2A_2e^{\zeta_2}}{1 + B_2e^{\zeta_2} + B_{22}e^{2\zeta_2}} = \frac{-2(p_2 + q_2)^2}{(-p_2 + q_2 + 2a) + 2\sqrt{(-p_2 + a)(q_2 + a)}\cosh(\zeta_2 - \theta_2)}, \\
\exp(\theta_2) &= (p_2 + q_2) \sqrt{\frac{q_2 + a}{-p_2 + a}}.
\end{aligned}$$

(2) After collision ($y \rightarrow +\infty$)

Soliton 1 ($\zeta_1 \approx 0, \zeta_2 \rightarrow -\infty$):

$$\begin{aligned}
u &\rightarrow \frac{2A_1e^{\zeta_1}}{1 + B_1e^{\zeta_1} + B_{11}e^{2\zeta_1}} = \frac{-2(p_1 + q_1)^2}{(-p_1 + q_1 + 2a) + 2\sqrt{(-p_1 + a)(q_1 + a)}\cosh(\zeta_1 - \theta_3)}, \\
\exp(\theta_3) &= (p_1 + q_1) \sqrt{\frac{q_1 + a}{a - p_1}}.
\end{aligned}$$

Soliton 2 ($\zeta_2 \approx 0, \zeta_1 \rightarrow +\infty$):

$$\begin{aligned}
u &\rightarrow \frac{2A_{112}e^{\zeta_2}}{B_{11} + B_{112}e^{\zeta_2} + B_{1122}e^{2\zeta_2}} = \frac{-2(p_2 + q_2)^2}{(-p_2 + q_2 + 2a) + \sqrt{(-p_2 + a)(q_2 + a)}\cosh(\zeta_2 - \theta_4)}, \\
\exp(\theta_4) &= \frac{(p_2 + q_1)(p_1 + q_2)(p_2 + q_2)}{(p_1 - p_2)(q_1 - q_2)} \sqrt{\frac{q_2 + a}{-p_2 + a}}.
\end{aligned}$$

The asymptotic analysis shows that the collision of two solitons is completely elastic. In other words, shapes and velocities of two solitary waves keep unchanged after the collision. The collision produces phase shifts θ for Soliton 1 and $-\theta$ for Soliton 2, where $\exp(\theta) = \frac{(p_2 + q_1)(p_1 + q_2)}{(p_1 - p_2)(q_1 - q_2)}$. The plots of the interactions of two solitons together with the corresponding density on the zero background are depicted in Fig. 3, Fig. 4 and Fig. 5. In Fig. 3, it exhibits the interaction of two bright solitons. Fig. 4 displays the interaction of two solitons with one above the horizon and the other under the horizon. Fig. 5 shows the interaction between two solitons that are both under the horizon.

3.3. Two-soliton solutions on the periodic background

To construct two-soliton solutions on the periodic background, we take $N = 3$ and $q_3 = -p_3^*$ in (13)–(15). The other parameters are chosen similar to those in Section 3.2. The tau functions f and g have the determinant expressions

$$f = \det(m_{ij}^{(1)})_{1 \leq i, j \leq 3}, \quad g = \det(m_{ij}^{(0)})_{1 \leq i, j \leq 3}, \quad (32)$$

where the entries are defined by

$$m_{ij}^{(n)} = c_j \delta_{i,j} + \frac{1}{p_i + q_j} \left(-\frac{p_i - a}{q_j + a} \right)^n e^{\xi_i + \eta_j}, \quad n = 0, 1, \quad (33)$$

with the parameters

$$p_3 = \alpha + \beta i, \quad q_3 = -p_3^*, \quad c_1 = c_2 = 1, \quad c_3 = ci, \quad \alpha = a, \quad (34)$$

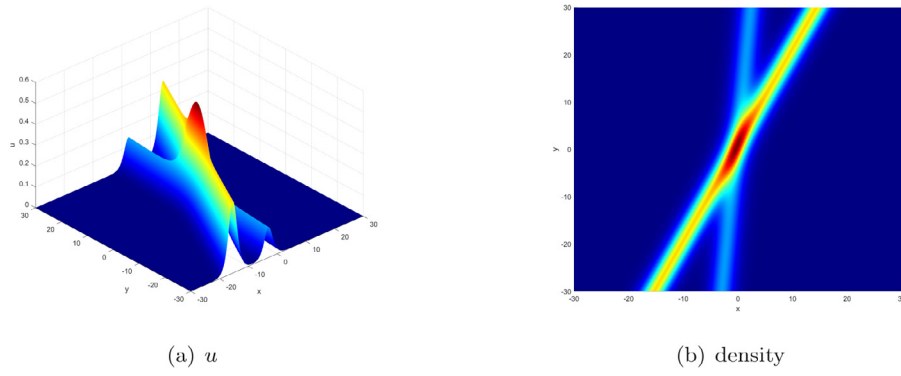


Fig. 3. Two-soliton solution u at $t = 0$ and the corresponding density plot with $a = 2, p_1 = 6, q_1 = -5, p_2 = 4, q_2 = -3, \xi_{i0} = \eta_{i0} = 0, c_1 = c_2 = 1$.

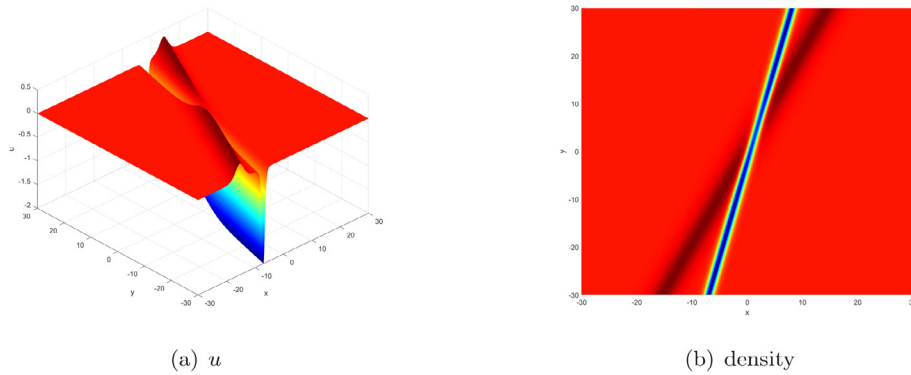


Fig. 4. Two-soliton solution u at $t = 0$ and the corresponding density plot with $a = 2, p_1 = 1, q_1 = 2, p_2 = 4, q_2 = -3, \xi_{i0} = \eta_{i0} = 0, c_1 = c_2 = 1$.

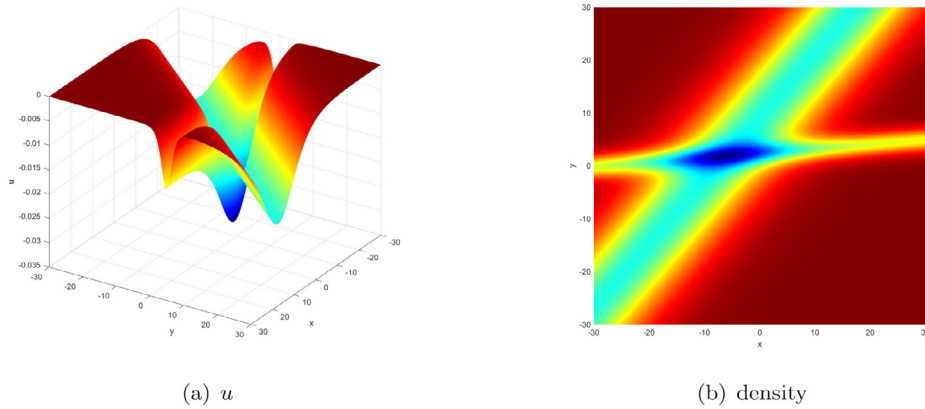


Fig. 5. Two-soliton solution u at $t = 0$ and the corresponding density plot with $a = 2, p_1 = \frac{7}{4}, q_1 = -\frac{5}{3}, p_2 = 1, q_2 = -\frac{4}{5}, \xi_{i0} = \eta_{i0} = 0, c_1 = c_2 = 1$.

and

$$\xi_i + \eta_i = (p_i + q_i)x + (q_i^2 - p_i^2)t + \left(\frac{1}{q_i + a} + \frac{1}{p_i - a}\right)y + \xi_{i0} + \eta_{i0}, \quad i = 1, 2, \quad (35)$$

$$\xi_3 + \eta_3 = 2(\beta x - 2\beta at - \frac{1}{\beta}y) + \xi_{30} + \eta_{30}. \quad (36)$$

The parameters p_1, p_2 and q_1, q_2, a, β, c are real numbers. Substituting f and g above into the expression (5) leads to the two-soliton solution on the periodic background. We plot the two-soliton solution on the periodic background in Fig. 6. The profile Fig. 6(a) and the density plot Fig. 6(b) show that when we set $p_3 = -q_3^*$ and take the same values of p_1, p_2, q_1, q_2, c as those in Section 3.2, the shape of waves becomes periodic instead of line smooth and the background becomes periodic instead of zero constant.

4. Line breather solutions

In this part, we will investigate dynamics of line breather solutions of DLW system.

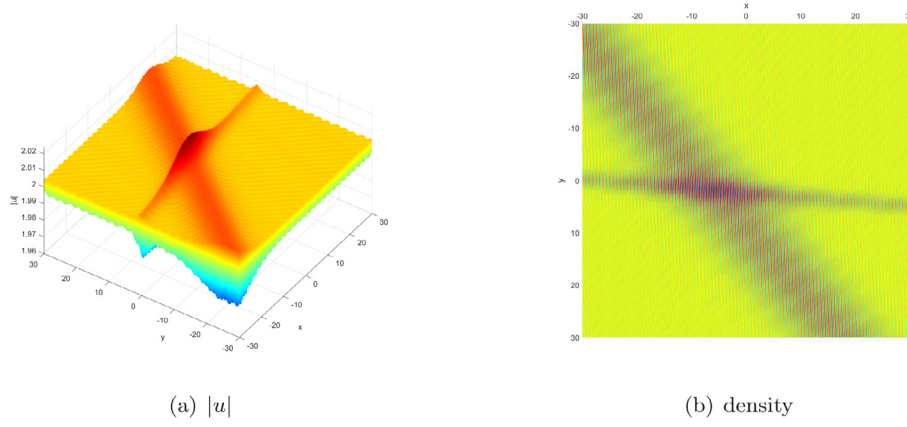


Fig. 6. Two-soliton solutions on the periodic background with $t = 0$, $a = 2$, $p_1 = \frac{7}{4}$, $q_1 = -\frac{5}{3}$, $p_2 = 1$, $q_2 = -\frac{4}{5}$, $p_3 = 2 + 5i$, $q_3 = -2 + 5i$, $c_1 = c_2 = 1$, $c_3 = 2i$, $\xi_{i0} = \eta_{i0} = 0$, $i = 1, 2, 3$.

4.1. Line breather on the constant background

To construct line breather solutions of the DLW system on the constant background, firstly, we set the size of the determinant to be even in [Theorem 2.1](#). We choose parameters $\tilde{p}_i, \tilde{q}_i, (i = 1, 2, \dots, N)$ to be purely imaginary and $\tilde{p}_i \neq -\tilde{q}_i$. For simplicity, we suppose $N = 2$, $\tilde{p}_1 = \tilde{q}_2 = pi$, $\tilde{q}_1 = \tilde{p}_2 = qi$, $c_1 = \alpha + \beta i$, $c_2 = -c_1^*$, p, q, α, β are real and $p \neq -q$, $\beta \neq 0$. In this case, the solutions u and v in [Theorem 2.1](#) are expressed by

$$u = 2 \ln \left(\frac{f}{g} \right)_x, \quad v = 2 \ln \left(fg \right)_{xy},$$

where

$$\begin{aligned} f &= \begin{vmatrix} (\alpha + \beta i)e^{-\xi_1 - \eta_1} - \frac{pi}{qi} \frac{1}{pi + qi} & (-\frac{pi}{pi}) \frac{1}{pi + pi} \\ (-\frac{qi}{qi}) \frac{1}{qi + qi} & (-\alpha + \beta i)e^{-\xi_2 - \eta_2} - \frac{qi}{pi} \frac{1}{qi + pi} \end{vmatrix} \\ &= -(\alpha^2 + \beta^2)e^{-\xi_1 - \eta_1 - \xi_2 - \eta_2} \left(1 - \frac{p}{q} b_1 e^{\xi_1 + \eta_1} - \frac{q}{p} b_2 e^{\xi_2 + \eta_2} + b_{12} e^{\xi_1 + \eta_1 + \xi_2 + \eta_2} \right), \\ g &= \begin{vmatrix} (\alpha + \beta i)e^{-\xi_1 - \eta_1} + \frac{1}{pi + qi} & \frac{1}{pi + pi} \\ \frac{1}{qi + qi} & (-\alpha + \beta i)e^{-\xi_2 - \eta_2} + \frac{1}{qi + pi} \end{vmatrix} \\ &= -(\alpha^2 + \beta^2)e^{-\xi_1 - \eta_1 - \xi_2 - \eta_2} \left(1 + b_1 e^{\xi_1 + \eta_1} + b_2 e^{\xi_2 + \eta_2} + b_{12} e^{\xi_1 + \eta_1 + \xi_2 + \eta_2} \right), \end{aligned}$$

and

$$\begin{aligned} \xi_1 + \eta_1 &= (p + q) \left(ix + (p - q - 2ai)t - \frac{i}{pq}y \right) + \xi_{10} + \eta_{10}, \\ \xi_2 + \eta_2 &= (p + q) \left(ix + (q - p - 2ai)t - \frac{i}{pq}y \right) + \xi_{20} + \eta_{20}, \\ b_1 &= \frac{1}{(p + q)(-\beta + \alpha i)}, \quad b_2 = \frac{1}{(p + q)(-\beta - \alpha i)}, \quad b_{12} = -\frac{(p - q)^2}{4pq(p + q)^2(\alpha^2 + \beta^2)}. \end{aligned}$$

It is clear that the solutions u and v are periodic on x, y and t and they display line breathers. The solution u is regular when $fg \neq 0$, which is equivalent to

$$F \equiv 1 - \frac{p}{q} b_1 e^{\xi_1 + \eta_1} - \frac{p}{q} b_2 e^{\xi_2 + \eta_2} + b_{12} e^{\xi_1 + \eta_1 + \xi_2 + \eta_2} \neq 0, \quad (37)$$

$$G \equiv 1 + b_1 e^{\xi_1 + \eta_1} + b_2 e^{\xi_2 + \eta_2} + b_{12} e^{\xi_1 + \eta_1 + \xi_2 + \eta_2} \neq 0. \quad (38)$$

If we take $\alpha = 0$ and $\xi_{10} + \eta_{10} = \xi_{20} + \eta_{20} = \zeta_0$, $pq > 0$, then the condition $F \neq 0$ is equivalent to

$$2b_{12}^2 e^{2\zeta_0} \neq \frac{p^2 b^2}{q^2} - 2b_{12} \pm \frac{pb}{q} \left(\frac{p^2 b^2}{q^2} - 4b_{12} \right)^{1/2}, \quad (39)$$

and $G \neq 0$ is equivalent to

$$2b_{12}^2 e^{2\zeta_0} \neq b^2 - 2b_{12} \pm b \left(b^2 - 4b_{12} \right)^{1/2}, \quad (40)$$

with $b = \frac{-1}{(p+q)\beta} (e^{(p-q)t} + e^{(q-p)t})$. The proof is similar to that in [\[16\]](#) and one can find details therein. To illustrate the dynamics of the breather solution u , we take the parameters

$$\alpha = 0, \beta = -1, p = \frac{1}{10}, q = \frac{2}{5}, t = 0, a = 1, \quad (41)$$

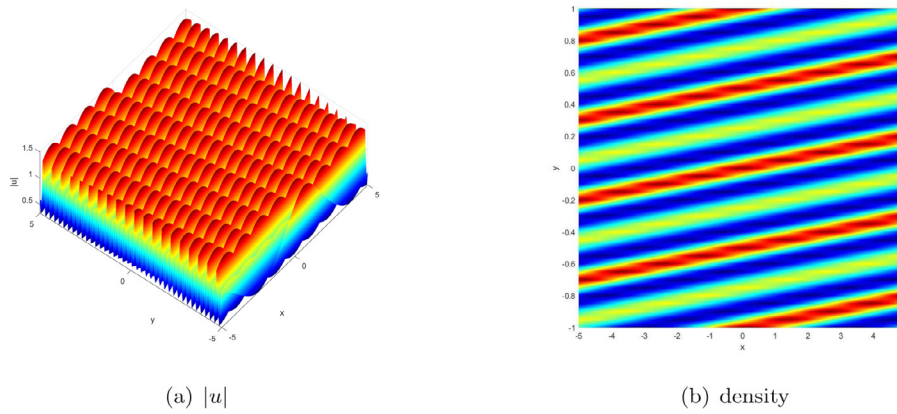


Fig. 7. Line breather solution on the constant background and the corresponding density plot. Parameters: $\alpha = 0, \beta = -1, p = \frac{1}{10}, q = \frac{2}{5}, t = 0, a = 1$.

and the line breather solution is expressed as

$$u = 50 \frac{4e^{-\frac{3}{2}i(x-25y)} + 9e^{-1/2i(x-25y)}}{(-4e^{-i(x-25y)} + 34e^{-\frac{1}{2}i(x-25y)} + 9)(4e^{-i(x-25y)} + 16e^{-\frac{1}{2}i(x-25y)} - 9)}. \quad (42)$$

Line breather solution u on the constant background is depicted in Figs. 7(a)–7(b). The periods of the solution (42) are 4π in x and $4\pi/25$ in y and the maximum and minimum values of $|u|$ under the parameters are approximately 1.5152 and 0.3331, respectively.

4.2. Line breather on the periodic background

To derive line breather solution on the periodic background, the size of the determinant in Theorem 2.1 should be odd, which is similar to the construction of two-soliton solution on the periodic background in Section 3.3. For simplicity, we suppose the determinant size is $N = 3$. By choosing $\tilde{p}_i, \tilde{q}_i (i, j = 1, 2, 3)$ to be purely imaginary and $\tilde{p}_1 = \tilde{q}_2 = pi, \tilde{q}_1 = \tilde{p}_2 = qi, \tilde{p}_3 = \tilde{q}_3 = mi$, we get the line breather solution on the periodic background with tau functions

$$f = \begin{vmatrix} (\alpha + \beta i)e^{-\xi_1 - \eta_1} - \frac{p}{q} \frac{1}{pi+qi} & -\frac{1}{2pi} & (-\frac{p}{m}) \frac{1}{pi+mi} \\ -\frac{1}{2qi} & (-\alpha + \beta i)e^{-\xi_2 - \eta_2} - \frac{q}{p} \frac{1}{qi+pi} & (-\frac{q}{m}) \frac{1}{qi+mi} \\ (-\frac{m}{q}) \frac{1}{mi+qi} & (-\frac{m}{p}) \frac{1}{mi+pi} & (-\alpha + \beta i)e^{-\xi_3 - \eta_3} - \frac{1}{2mi} \end{vmatrix}, \quad (43)$$

$$g = \begin{vmatrix} (\alpha + \beta i)e^{-\xi_1 - \eta_1} + \frac{1}{pi+qi} & \frac{1}{2pi} & \frac{1}{pi+mi} \\ \frac{1}{2qi} & (-\alpha + \beta i)e^{-\xi_2 - \eta_2} + \frac{1}{qi+pi} & \frac{1}{qi+mi} \\ \frac{1}{mi+qi} & \frac{1}{mi+pi} & (-\alpha + \beta i)e^{-\xi_3 - \eta_3} + \frac{1}{2mi} \end{vmatrix}. \quad (44)$$

Furthermore, we put the parameters $m = 1, c_3 = -i$ and the others have the values in (41). Then we obtain the line breather solution on the periodic background

$$u = 2 \ln \left(\frac{f}{g} \right)_x = 154 \frac{A}{BC},$$

with

$$\begin{aligned} A &= 786671424e^{-3i(x-9y)} - 123268068e^{-\frac{1}{2}i(5x-29y)} - 36374184e^{-2i(x-y)} \\ &\quad + 4428675e^{-\frac{1}{2}i(x-25y)} - 58436224e^{-4i(x-13y)} + 16008300e^{-\frac{3}{2}i(x-25y)} \\ &\quad - 182613200e^{-\frac{1}{2}i(11x-83y)} + 156241008e^{-\frac{1}{2}i(7x-79y)} - 410879700e^{-\frac{3}{2}i(3x-11y)}, \\ B &= 47432e^{-3i(x-9y)} - 23716e^{-i(x-25y)} - 403172e^{-\frac{1}{2}i(5x-29y)} + 70686e^{-\frac{1}{2}i(x-25y)} - 106722e^{-2i(x-y)} + 6561, \\ C &= 47432e^{-3i(x-9y)} + 23716e^{-i(x-25y)} + 189728e^{-\frac{1}{2}i(5x-29y)} + 33264e^{-\frac{1}{2}i(x-25y)} - 106722e^{-2i(x-y)} - 6561. \end{aligned}$$

The amplitudes of the periodic background (see Figs. 8(a)–8(b)) are higher than those of the constant background (see Figs. 7(a)–7(b)). The maximum and minimum values of $|u|$ on the periodic background are approximately 4.8409 and 0.3160, respectively.

5. Dynamics of the rational solutions

5.1. Fundamental algebraic solitons

In this part, we take $N = 1, n = 0, c_1 = 0$ in Theorem 2.1. In this case, DLW system (1)–(2) owns the solution (5) with

$$f = \tilde{m}_{11}^{(1)} = -\frac{\tilde{p}_1}{\tilde{q}_1} \frac{1}{\tilde{p}_1 + \tilde{q}_1} \left[(1 + \tilde{\xi}_1 + c_{11} - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})(-1 + \tilde{\eta}_1 + d_{11} - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1}) + \frac{\tilde{p}_1 \tilde{q}_1}{(\tilde{p}_1 + \tilde{q}_1)^2} \right], \quad (45)$$

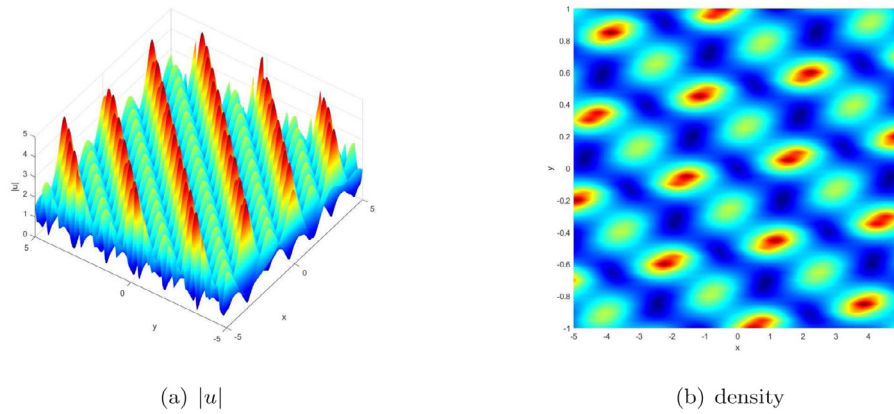


Fig. 8. Line breather solutions on the periodic background with parameters $\alpha = 0, \beta = -1, p = \frac{1}{10}, q = \frac{2}{5}, t = 0, a = 1, \tilde{p}_3 = \tilde{q}_3 = i, c_3 = -i$ and the corresponding density plot.

$$g = \tilde{m}_{11}^{(0)} = \frac{1}{\tilde{p}_1 + \tilde{q}_1} \left[(\tilde{\xi}_1 + c_{11} - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})(\tilde{\eta}_1 + d_{11} - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1}) + \frac{\tilde{p}_1 \tilde{q}_1}{(\tilde{p}_1 + \tilde{q}_1)^2} \right], \quad (46)$$

where

$$\tilde{\xi}_1 = \tilde{p}_1 x - 2\tilde{p}_1(\tilde{p}_1 + a)t - \frac{1}{\tilde{p}_1}y, \quad \tilde{\eta}_1 = \tilde{q}_1 x + 2\tilde{q}_1(\tilde{q}_1 - a)t - \frac{1}{\tilde{q}_1}y.$$

Here we choose parameters

$$c_{11} = ci, \quad d_{11} = di, \quad (47)$$

and set $\tilde{p}_1, \tilde{q}_1, a$ as real. Then

$$\begin{aligned} |f| &= \frac{\tilde{p}_1}{\tilde{q}_1} \frac{1}{\tilde{p}_1 + \tilde{q}_1} \left| \left(1 + \tilde{\xi}_1 + ci - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1} \right) \left(-1 + \tilde{\eta}_1 + di - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1} \right) + \frac{\tilde{p}_1 \tilde{q}_1}{(\tilde{p}_1 + \tilde{q}_1)^2} \right| \\ &= \frac{1}{\tilde{p}_1 + \tilde{q}_1} \sqrt{I_{11}^2 + I_{12}^2}, \\ I_{11} &= (\tilde{\xi}_1 + 1 - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})(\tilde{\eta}_1 - 1 - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1}) - cd + \frac{\tilde{p}_1 \tilde{q}_1}{(\tilde{p}_1 + \tilde{q}_1)^2}, \\ I_{12} &= (\tilde{\xi}_1 + 1 - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})d + (\tilde{\eta}_1 - 1 - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1})c, \end{aligned}$$

and

$$\begin{aligned} |g| &= \frac{1}{\tilde{p}_1 + \tilde{q}_1} \sqrt{I_{21}^2 + I_{22}^2}, \\ I_{21} &= (\tilde{\xi}_1 - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})(\tilde{\eta}_1 - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1}) - cd + \frac{\tilde{p}_1 \tilde{q}_1}{(\tilde{p}_1 + \tilde{q}_1)^2}, \\ I_{22} &= (\tilde{\xi}_1 - \frac{\tilde{p}_1}{\tilde{p}_1 + \tilde{q}_1})d + (\tilde{\eta}_1 - \frac{\tilde{q}_1}{\tilde{p}_1 + \tilde{q}_1})c. \end{aligned}$$

To ensure the regularity of $|u|$ and $|v|$, we need to take parameters so that $I_{11}^2 + I_{12}^2 \neq 0$ and $I_{21}^2 + I_{22}^2 \neq 0$.

Collision properties of algebraic solitons for nonlocal NLS equations were discussed in [21,22]. We analyze the collision of the two-algebraic-soliton solution for DLW system. For convenience, we assume $t = 0$ and denote $\tilde{p}_1 = p, \tilde{q}_1 = q$. Then we have

$$\tilde{\xi}_1 = \tilde{p}_1 x - \frac{1}{\tilde{p}_1}y = \tilde{p}_1(x - \frac{1}{\tilde{p}_1^2}y) = p\xi, \quad \tilde{\eta}_1 = \tilde{q}_1 x - \frac{1}{\tilde{q}_1}y = \tilde{q}_1(x - \frac{1}{\tilde{q}_1^2}y) = q\eta,$$

where $\xi = x - \frac{1}{p^2}y, \eta = x - \frac{1}{q^2}y$. The solution v is expressed by

$$\begin{aligned} v &= 2(\ln fg)_{xy} = 2 \left(\frac{f_{xy}}{f} - \frac{f_x f_y}{f^2} + \frac{g_{xy}}{g} - \frac{g_x g_y}{g^2} \right), \\ f_x &= -\frac{p}{q} \frac{1}{p+q} \left[q(1 + p\xi + c_{11} - \frac{p}{p+q}) + p(-1 + q\eta + d_{11} - \frac{q}{p+q}) \right], \\ f_y &= -\frac{p}{q} \frac{1}{p+q} \left[-\frac{1}{q}(1 + p\xi + c_{11} - \frac{p}{p+q}) - \frac{1}{p}(-1 + q\eta + d_{11} - \frac{q}{p+q}) \right], \\ g_x &= \frac{1}{p+q} \left[q(p\xi + c_{11} - \frac{p}{p+q}) + p(q\eta + d_{11} - \frac{q}{p+q}) \right], \end{aligned}$$

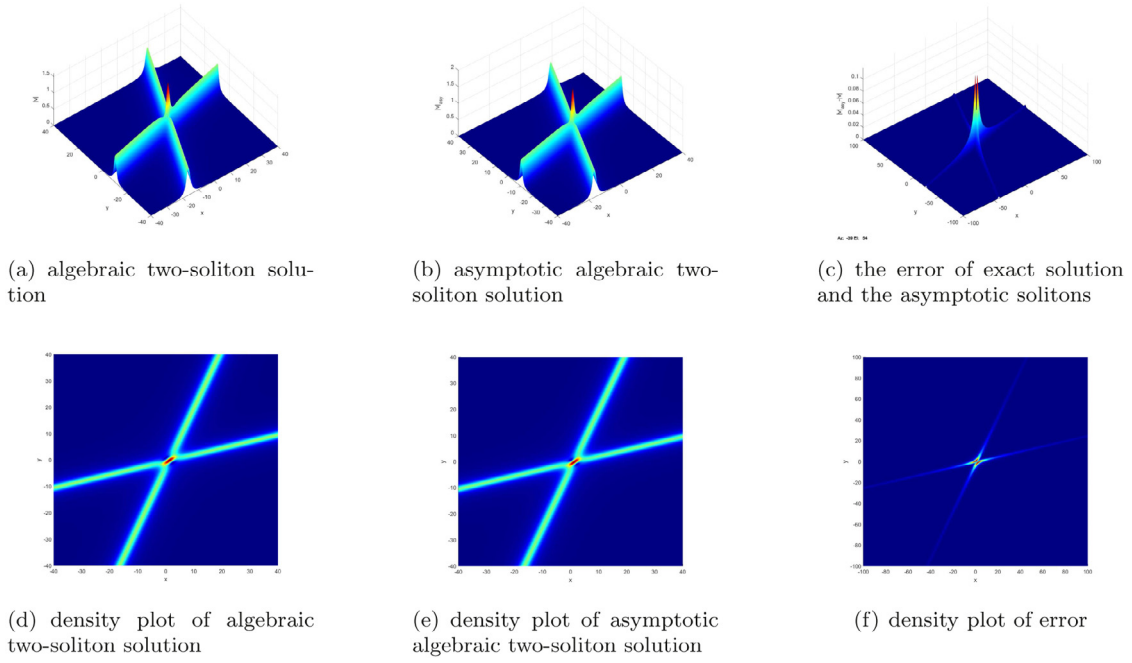


Fig. 9. The exact algebraic soliton solution v and the asymptotic solitons v_{asy} at $t = 0$ with parameters $\tilde{p}_1 = -\frac{1}{2}$, $\tilde{q}_1 = \frac{3}{2}$, $c = d = 2$, $a = 1$.

$$g_y = \frac{1}{p+q} \left[-\frac{1}{q} \left(p\xi + c_{11} - \frac{p}{p+q} \right) - \frac{1}{p} \left(q\eta + d_{11} - \frac{q}{p+q} \right) \right],$$

$$f_{xy} = -\frac{p}{q} \frac{1}{p+q} \left(-\frac{p}{q} - \frac{q}{p} \right), \quad g_{xy} = -\frac{p}{q} - \frac{q}{p}.$$

Obviously, v depends on ξ and η . Since $\xi - \eta = \frac{p^2 - q^2}{p^2 q^2} y$, the asymptotic solitons of v appear only on the following two asymptotic cases: (1) $|\xi| = O(|y|)$, $|\eta| = O(1)$, (2) $|\eta| = O(|y|)$, $|\xi| = O(1)$. In case (1), the explicit expression of v is given in the [Appendix](#). The Taylor expansion of v with ξ^{-1} yields

$$v = 2(p+q)^2 \left(\frac{1}{[(q\eta + d - 1)(p+q) - q]^2} + \frac{1}{[(q\eta + d)(p+q) - q]^2} \right) + O(\xi^{-1}) \equiv v_1. \quad (48)$$

Similarly, case (2) leads to

$$v = 2(p+q)^2 \left(\frac{1}{[(p\xi + c + 1)(p+q) - p]^2} + \frac{1}{[(p\xi + c)(p+q) - p]^2} \right) + O(\eta^{-1}) \equiv v_2, \quad (49)$$

where $O(\xi^{-1})$ and $O(\eta^{-1})$ can be replaced by $O(y^{-1})$ through the asymptotic conditions. From the asymptotic analysis above, the solution v approaches to two asymptotic solitons v_1 and v_2 in two cases with the rate $O(t^{-1})$, respectively. To give exact expression of algebraic two-soliton solution, firstly, we consider $\tilde{p}_1 \neq \tilde{q}_1$ and take

$$\tilde{p}_1 = -\frac{1}{2}, \quad \tilde{q}_1 = \frac{3}{2}, \quad c = d = 2, \quad a = 1, \quad t = 0. \quad (50)$$

In this case, from (45)–(5) and (46), solution v has the expression

$$\begin{aligned} v = & 32(-485352 - 967680iyx + 806400ixy^2 - 30720ixy^3 + 231552ix^2y - 894240ix - 528768iy - 1069056iy^2 \\ & - 165888ix^2 + 55296iy^3 + 241056ix^3 + 1173312y^2 - 334368yx - 93960ix^4 + 1033236x^2 + 194560iy^4 + 48384ix^2y^2 \\ & - 21504ix^2y^3 + 231552ix^3y - 47520ix^4y - 16128ix^3y^2 - 112640ixy^4 + 13608ix^5 + 57344iy^5 + 3645x^6 \\ & + 20480y^6 - 34506x^5 + 174080y^5 + 173178x^4 + 670208y^4 - 38232x^5y + 143280x^4y^2 - 251136x^3y^3 + 254720x^2y^4 \\ & - 120832xy^5 + 292680x^4y - 857664x^3y^2 + 1132800x^2y^3 - 750080xy^4 - 1111968x^3y + 1796544x^2y^2 - 2087424xy^3 \\ & - 1817856y^2x + 2130624yx^2 + 1422336y^3 - 511920x^3 + 899424i - 876744x - 1200096y) / \\ & [(24ix + 32iy + 9x^2 - 40yx + 16y^2 + 102 - 24i - 42x + 72y)^2(24ix + 32iy + 9x^2 - 40yx + 16y^2 + 66 - 24i - 18x + 40y)^2]. \end{aligned}$$

Fig. 9 describes exact and asymptotic algebraic two-soliton solution of v together with the error. Here we choose $v_{asy} = v_1 + v_2$. One can see that the collision is completely elastic in **Fig. 9(a)**. There are one peak and two troughs under the wave plane. The similar rational solutions were found in the study of 2+1-dimensional sinh-Gordon equation [39].

We find that the sign of $\tilde{q}_1(\tilde{q}_1 - a)$ determines propagation speed and direction of algebraic solitons. When we take a new set of parameters

$$\tilde{p}_1 = \tilde{q}_1 = 1, \quad c = 2, \quad d = 3, \quad a = 2, \quad (51)$$

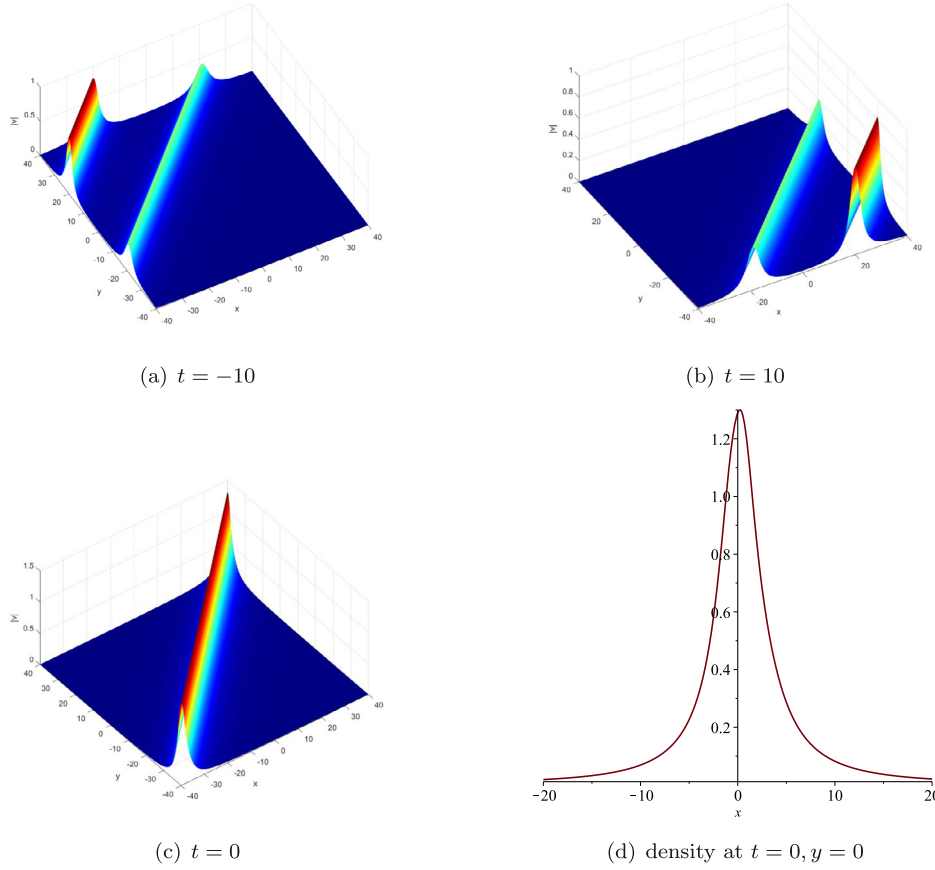


Fig. 10. The algebraic soliton solution v at different time with parameters $c = 2$, $d = 3$, $\tilde{p}_1 = \tilde{q}_1 = 1$, $a = 1$.

two algebraic solitons become parallel and the higher one owns fast speed so that it catches up with the lower one and then two solitons separate each other after the collision (see Fig. 10). When $y = t = 0$, the parallel two-soliton solution v has simple form

$$v = 64 \frac{30ix^5 + 2x^6 - 74ix^4 - 6x^5 - 552ix^3 - 180x^4 + 861ix^2 + 360x^3 + 755ix + 908x^2 - 247 - 462i - 1009x}{(-13 - 3i + 10ix + 2x^2 - 2x)^2 (-11 - 5i + 10ix + 2x^2 - 2x)^2}.$$

and the plot is displayed in Fig. 10(d). It is easy to verify the regularity of the solution. We see that v decays algebraically to the wave plane. When parameters satisfy $\tilde{p}_1 = \tilde{q}_1 = a$, one can find that one soliton propagates regularly and the other one is stationary.

Since there is an arbitrary parameter a in DLW system, it is natural to consider the influence of the constant. Figs. 11(a)–11(d) display the algebraic soliton solutions v with different values of a . As we can see, parameter a affects the direction of wave propagation and amplitudes. The two waves are almost parallel each other at $a = -3$ and $a = 5$, while almost perpendicular when $a = 0.83$.

5.2. Algebraic multi-soliton solutions

In this part, we assume the determinant size $N > 1$ to derive algebraic multi-soliton solutions. In particular, we take $N = 2$, $n_1 = 1$, $n = 0$, $c_1 = 0$ in Theorem 2.1. Thus we obtain

$$f = \begin{vmatrix} \tilde{m}_{11}^{(1)} & \tilde{m}_{12}^{(1)} \\ \tilde{m}_{21}^{(1)} & \tilde{m}_{22}^{(1)} \end{vmatrix}, g = \begin{vmatrix} \tilde{m}_{11}^{(0)} & \tilde{m}_{12}^{(0)} \\ \tilde{m}_{21}^{(0)} & \tilde{m}_{22}^{(0)} \end{vmatrix}, \quad (52)$$

where

$$\tilde{m}_{ij}^{(1)} = -\frac{\tilde{p}_i}{\tilde{q}_j} \frac{1}{\tilde{p}_i + \tilde{q}_j} \left[(1 + \tilde{\xi}_i + c_{i1} - \frac{\tilde{p}_i}{\tilde{p}_i + \tilde{q}_j})(-1 + \tilde{\eta}_j + d_{j1} - \frac{\tilde{q}_j}{\tilde{p}_i + \tilde{q}_j}) + \frac{\tilde{p}_i \tilde{q}_j}{(\tilde{p}_i + \tilde{q}_j)^2} \right], \quad (53)$$

$$\tilde{m}_{ij}^{(0)} = \frac{1}{\tilde{p}_i + \tilde{q}_j} \left[(\tilde{\xi}_i + c_{i1} - \frac{\tilde{p}_i}{\tilde{p}_i + \tilde{q}_j})(\tilde{\eta}_j + d_{j1} - \frac{\tilde{q}_j}{\tilde{p}_i + \tilde{q}_j}) + \frac{\tilde{p}_i \tilde{q}_j}{(\tilde{p}_i + \tilde{q}_j)^2} \right], \quad (54)$$

and

$$\tilde{\xi}_i = \tilde{p}_i x - 2\tilde{p}_i(\tilde{p}_i + a)t - \frac{1}{\tilde{p}_i} y, \quad \tilde{\eta}_j = \tilde{q}_j x + 2\tilde{q}_j(\tilde{q}_j - a)t - \frac{1}{\tilde{q}_j} y.$$

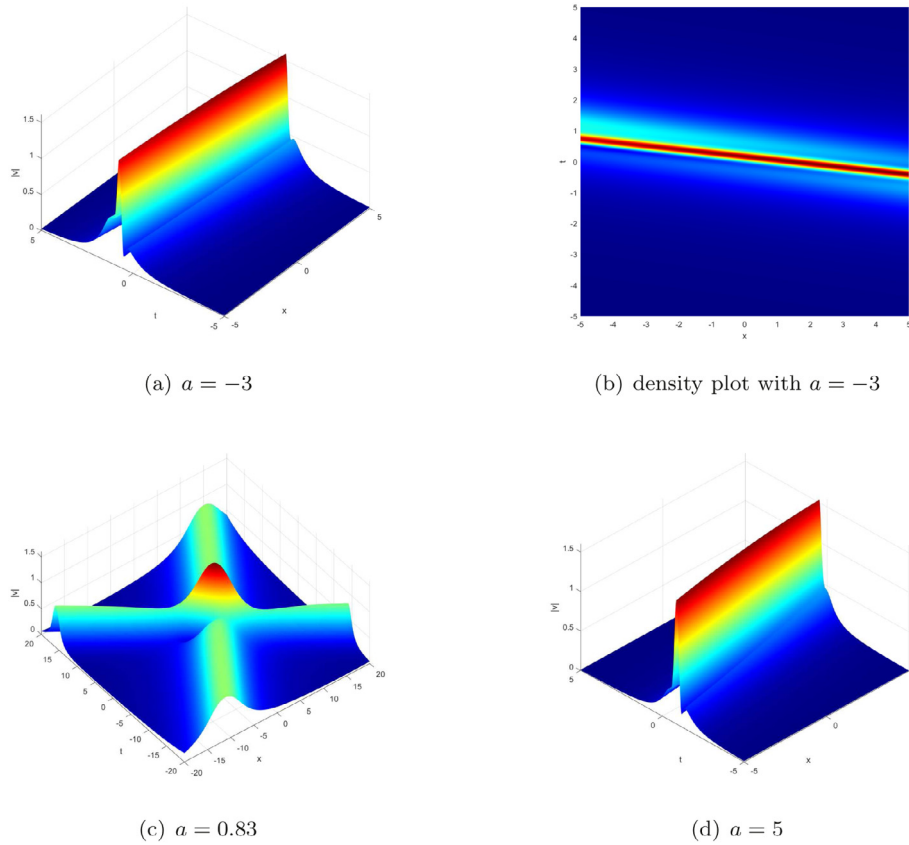


Fig. 11. The algebraic soliton solution v with different a with parameters $\tilde{p}_1 = -\frac{1}{2}$, $\tilde{q}_1 = \frac{3}{2}$, $c = d = 2$, $y = 0$.

To illustrate the dynamics of the solution v , we first choose parameters

$$\tilde{p}_1 = -\frac{1}{2}, \tilde{q}_1 = \frac{3}{2}, \tilde{p}_2 = -\frac{5}{2}, \tilde{q}_2 = 2, a = 1, c_{11} = 2i, c_{21} = i, d_{11} = 3i, d_{21} = 20i. \quad (55)$$

Figs. 12(a)–12(i) display the dynamics of potential v with parameters (55). There are three waves with different propagation direction and different speeds on the wave plane. At different time t , the position of the collision center changes (see Fig. 12(b), 12(e), 12(h)). The 2-dimensional images Figs. 12(c), 12(f) and 12(i) show that waves with higher amplitudes catch up with lower waves. After collision, waves separate each other with some change of heights.

If we choose parameters as

$$\tilde{p}_1 = \tilde{q}_2 = -\frac{1}{2}, \tilde{p}_2 = \tilde{q}_1 = \frac{3}{2}, a = 1, c_{11} = 2i, c_{21} = 4i, d_{11} = 2i, d_{21} = 10i, \quad (56)$$

the image of the potential v degenerates into that of algebraic two-soliton solution in Section 5.2. Plot of the degeneration algebraic two-soliton v is depicted in Fig. 13(a).

5.3. Multiple rational solutions on periodic background

In order to construct multiple rational solutions on the periodic background, we choose the Gram determinant size $N = 3$. In this case, f , g are expressed by

$$f = \begin{vmatrix} \tilde{m}_{11}^{(1)} & \tilde{m}_{12}^{(1)} & \tilde{m}_{13}^{(1)} \\ \tilde{m}_{21}^{(1)} & \tilde{m}_{22}^{(1)} & \tilde{m}_{23}^{(1)} \\ \tilde{m}_{31}^{(1)} & \tilde{m}_{32}^{(1)} & \tilde{m}_{33}^{(1)} \end{vmatrix}, \quad g = \begin{vmatrix} \tilde{m}_{11}^{(0)} & \tilde{m}_{12}^{(0)} & \tilde{m}_{13}^{(0)} \\ \tilde{m}_{21}^{(0)} & \tilde{m}_{22}^{(0)} & \tilde{m}_{23}^{(0)} \\ \tilde{m}_{31}^{(0)} & \tilde{m}_{32}^{(0)} & \tilde{m}_{33}^{(0)} \end{vmatrix}, \quad (57)$$

where $\tilde{m}_{ij}^{(n)}$ ($n = 0, 1$, $i, j = 1, 2$) are given as (53)–(54) and

$$\begin{cases} \tilde{m}_{ij}^{(n)} = \left(-\frac{\tilde{p}_i}{\tilde{q}_j}\right)^n \frac{1}{\tilde{p}_i + \tilde{q}_j} (n + \tilde{\xi}_i + c_{i1} - \frac{\tilde{p}_i}{\tilde{p}_i + \tilde{q}_j}), i = 1, 2; j = 3 \\ \tilde{m}_{ij}^{(n)} = \left(-\frac{\tilde{p}_i}{\tilde{q}_j}\right)^n \frac{1}{\tilde{p}_i + \tilde{q}_j} (-n + \tilde{\eta}_j + d_{j1} - \frac{\tilde{q}_j}{\tilde{p}_i + \tilde{q}_j}), j = 1, 2; i = 3 \\ \tilde{m}_{ij}^{(n)} = c_j e^{-\xi_i - \eta_j} + \left(-\frac{\tilde{p}_i}{\tilde{q}_j}\right)^n \frac{1}{\tilde{p}_i + \tilde{q}_j}, i = j = 3 \end{cases}$$

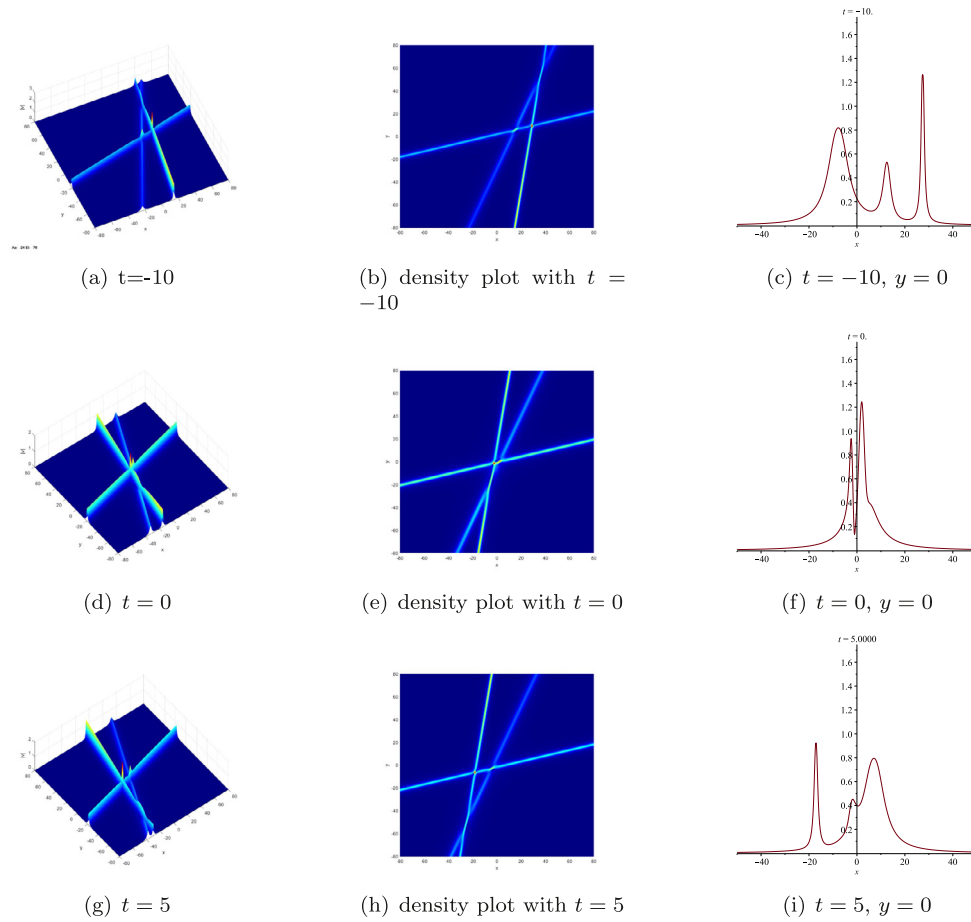


Fig. 12. The algebraic three-soliton solution v at different time with parameters $\tilde{p}_1 = -\frac{1}{2}$, $\tilde{q}_1 = \frac{3}{2}$, $\tilde{p}_2 = -\frac{5}{2}$, $\tilde{q}_2 = 2$, $a = 1$, $c_{11} = 2i$, $c_{21} = i$, $d_{11} = 3i$, $d_{21} = 20i$.

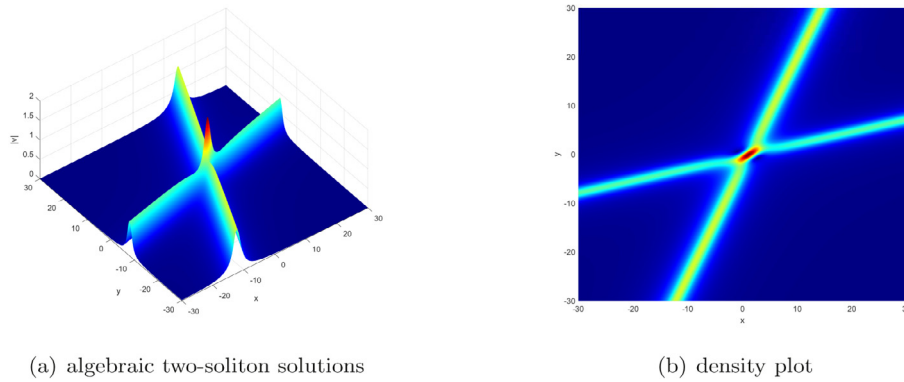


Fig. 13. The algebraic soliton solution v at $t = 0$ with parameters $\tilde{p}_1 = \tilde{q}_2 = -\frac{1}{2}$, $\tilde{p}_2 = \tilde{q}_1 = \frac{3}{2}$, $a = 1$, $c_{11} = 2i$, $c_{21} = 4i$, $d_{11} = 2i$, $d_{21} = 10i$.

Substituting f and g into Eq. (5) leads to the semi-rational solution u and v . Since the explicit expression for u and v is very complicated, we omit the exact form of u and v here. To study the dynamics of this kind of semi-rational solution, we put the parameters $\tilde{p}_3 = \tilde{q}_3 = -i$, $c_3 = -2i$ and the other parameters are the same as (55), Fig. 14(a) displays three periodic waves on the periodic background plane where two waves are anti-dark and the third one is dark. While we take parameters as (56), Fig. 15(a) shows two periodic wave on the periodic background that one wave is anti-dark and the other is dark. One can see that the amplitudes of the waves have increased after the collision and the propagation direction has no change.

5.4. Lump solutions for the intermediate potential

We know that the ratio of rational tau functions can generate lump solutions. Thus we consider the potential $W = \frac{f}{g}$ in the solution (52). In this case, we can obtain multiple lump solutions on the constant and periodic background (see Fig. 8, Fig. 9), respectively.

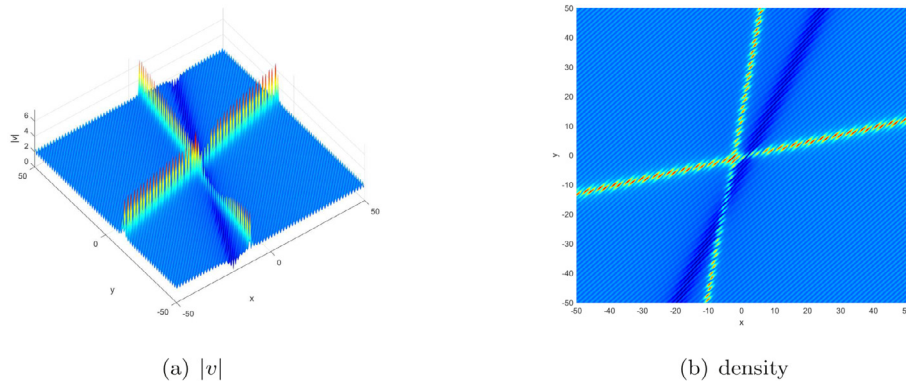


Fig. 14. The semi-rational solution v at $t = 0$ on the periodic background with parameters in (55) and $\tilde{p}_3 = \tilde{q}_3 = -i$, $c_3 = -2i$ and (b) is the corresponding density plot.

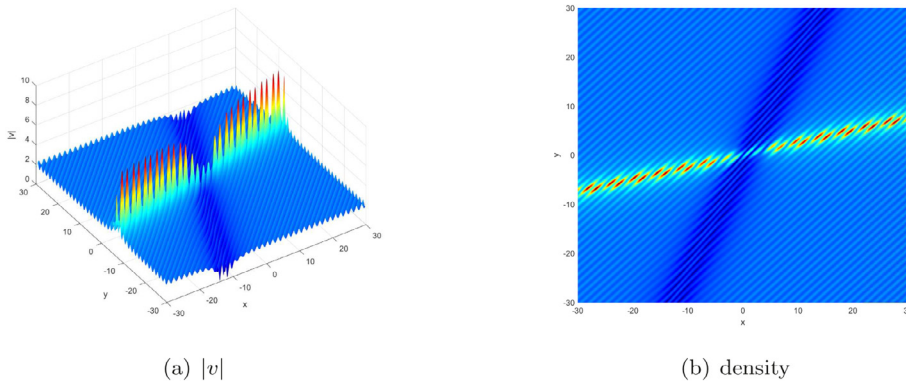


Fig. 15. The semi-rational solution v at $t = 0$ on the periodic background with parameters in (56) and $\tilde{p}_3 = \tilde{q}_3 = -i$, $c_3 = -2i$ and (b) is the corresponding density plot.

Actually, when we choose

$$\tilde{q}_i = \tilde{p}_i^*, \tilde{p}_1 + \tilde{q}_1 = -(\tilde{p}_2 + \tilde{q}_2), t = 0, c_{11} = c_{21} = d_{11} = d_{21} = 0, \quad (58)$$

the regularity can be verified similar to [16]. With parameters $\tilde{p}_1 = \frac{7}{10} - \frac{7}{10}i$, $\tilde{q}_1 = \frac{7}{10} + \frac{7}{10}i$, $\tilde{p}_2 = -\frac{7}{10} + \frac{7}{10}i$, $\tilde{q}_2 = -\frac{7}{10} - \frac{7}{10}i$, $t = 0$, the potential W can be rewritten as

$$W = -\frac{5764801x^4 + 12005000y^2x^2 + 6250000y^4 + 11764900ix^2 - 12250000iy^2 + 24010000yx - 6002500}{5764801x^4 + 12005000x^2y^2 + 6250000y^4 + 6002500}. \quad (59)$$

Since the denominator is positive, W is always regular. Fig. 16 displays four peaks and four troughs. With different imaginary parts of $\tilde{p}_i$, $\tilde{q}_j$, ($i, j = 1, 2$), the wave profiles change (see Figs. 16(a)–16(f)) accordingly. When $|p_i| = |q_i|$, the density plot Fig. 16(d) shows that the center part exhibits a circle. While $|p_i| \neq |q_i|$, the center part displays an ellipse, see Fig. 16(e) and Fig. 16(f).

In order to construct rational solutions of the potential W on the periodic background, we take $N = 3$ in Theorem 2.1. With parameters $c_3 = 5i$, $\tilde{p}_3 = \tilde{q}_3 = -\frac{3}{2}i$ and other parameters the same as those in Fig. 16(a), we obtain rational solution W on the periodic background. Fig. 17 shows that the wave plane becomes periodic and the shape of the wave is similar to the periodic solution depicted in Fig. 8(a).

5.5. High-order rational solutions

When $N = 1$, $n_1 > 1$, we obtain high-order rational solutions. Here we take $n = 0$, $n_1 = 2$ and tau functions are

$$f = -\frac{\tilde{p}_1}{\tilde{q}_1} \sum_{k=0}^2 c_{1k} (\tilde{p}_1 \partial_{\tilde{p}_1} + 1 + \tilde{\xi}_1)^{2-k} \times \sum_{l=0}^2 d_{1l} (\tilde{q}_1 \partial_{\tilde{q}_1} - 1 + \tilde{\eta}_1)^{2-l} \left(\frac{1}{\tilde{p}_1 + \tilde{q}_1} \right), \quad (60)$$

$$g = \sum_{k=0}^2 c_{1k} (\tilde{p}_1 \partial_{\tilde{p}_1} + \tilde{\xi}_1)^{2-k} \times \sum_{l=0}^2 d_{1l} (\tilde{q}_1 \partial_{\tilde{q}_1} + \tilde{\eta}_1)^{2-l} \left(\frac{1}{\tilde{p}_1 + \tilde{q}_1} \right). \quad (61)$$

Parameters

$$c_{10} = 1, \quad c_{11} = 0, \quad d_{10} = 1, \quad d_{11} = 0,$$

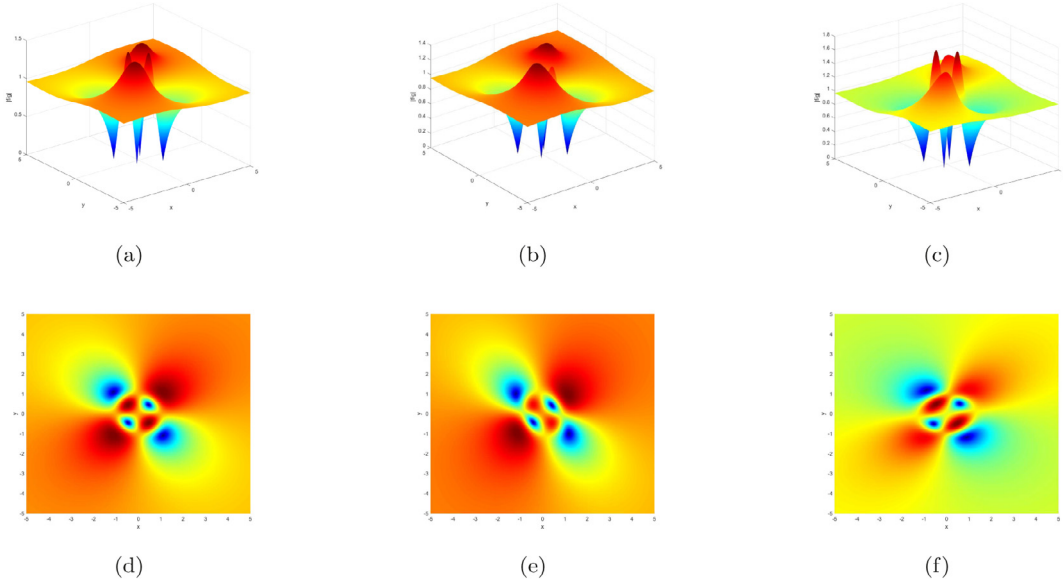


Fig. 16. Lump solutions and the corresponding density plots of the potential f/g on the constant background at $t = 0$. Parameters are given as $c_{11} = c_{21} = d_{11} = d_{21} = 0$ and (a) $\tilde{p}_1 = \tilde{q}_1^* = \frac{7}{10} - \frac{7}{10}i$, $\tilde{p}_2 = \tilde{q}_2^* = -\frac{7}{10} + \frac{7}{10}i$, (b) $\tilde{p}_1 = \tilde{q}_1^* = \frac{7}{10} - \frac{4}{5}i$, $\tilde{p}_2 = \tilde{q}_2^* = -\frac{7}{10} + \frac{4}{5}i$, (c) $\tilde{p}_1 = \tilde{q}_1^* = \frac{7}{10} - \frac{3}{5}i$, $\tilde{p}_2 = \tilde{q}_2^* = -\frac{7}{10} + \frac{3}{5}i$. (d), (e) and (f) are density plots corresponding to (a), (b), (c) respectively.

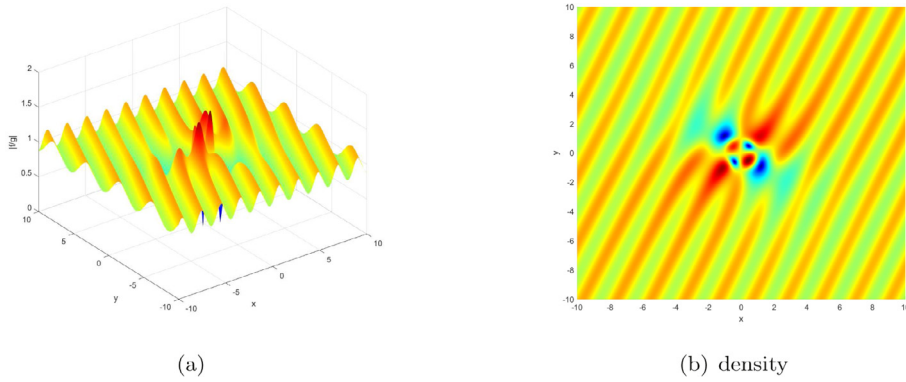


Fig. 17. Lump solutions and the corresponding density plot of the potential f/g on the periodic background with parameters $c_3 = 5i$, $\tilde{p}_3 = \tilde{q}_3 = -\frac{3}{2}i$, and other parameters are the same as those for Fig. 16(a).

lead to

$$f = -\frac{\tilde{p}_1}{\tilde{q}_1}((\tilde{p}_1\partial_{\tilde{p}_1} + 1 + \tilde{\xi}_1)^2 + c_{12}) \times ((\tilde{q}_1\partial_{\tilde{q}_1} - 1 + \tilde{\eta}_1)^2 + d_{12}) \left(\frac{1}{\tilde{p}_1 + \tilde{q}_1} \right), \quad (62)$$

$$g = ((\tilde{p}_1\partial_{\tilde{p}_1} + \tilde{\xi}_1)^2 + c_{12}) \times ((\tilde{q}_1\partial_{\tilde{q}_1} + \tilde{\eta}_1)^2 + d_{12}) \left(\frac{1}{\tilde{p}_1 + \tilde{q}_1} \right). \quad (63)$$

By setting the parameters c_{12} , d_{12} to be purely imaginary,

$$c_{12} = ci = 20i, \quad d_{12} = di = \pm 20i,$$

together with $\tilde{p}_1 = -\frac{1}{2}$, $\tilde{q}_1 = -2$, $a = 1$, we depict the image of $|v|$ at $t = 0$ in Figs. 18(a)–18(b). It is shown that there are two waves before collision (i.e. $y \rightarrow -\infty$), and four waves after collision (i.e. $y \rightarrow +\infty$). The wave heights tend to infinity when y approaches to positive infinity. When $d = 20$ (see Fig. 18(a)), one peak appears at the collision center. When $d = -20$ (see Fig. 18(c)), the collision center is sunken. The profiles are similar to those in [39].

Also we can construct high-order rational solution of the potential W . We depict Figs. 19(a)–19(b) with parameters $c = d = 0$, $a = 1$, $\tilde{p}_1 = 1 + i$, $\tilde{q}_1 = 1 - i$ for the potential W . In this case, the potential W at $t = 0$ is expressed as

$$\begin{aligned} W = & (-8y^2x + 48yx - 32x + 48x^2 - 16ix^4 - 8iy^2x^2 - iy^4 - 16yx^2 - 4y^3 + 4y^2 + 16ix^3 \\ & + 24iyx^2 + 20iy^2x - 2iy^3 - 32x^3 - 16y - 8iyx - 4iy^2 + 16 - 8ix) \\ & / (16x^4 + 8x^2y^2 + y^4 - 16x^3 - 24x^2y - 20xy^2 + 2y^3 + 32x^2 + 24xy + 12y^2 - 32x - 12y + 16). \end{aligned}$$

Fig. 19(b) shows that there are four parts at the collision center.

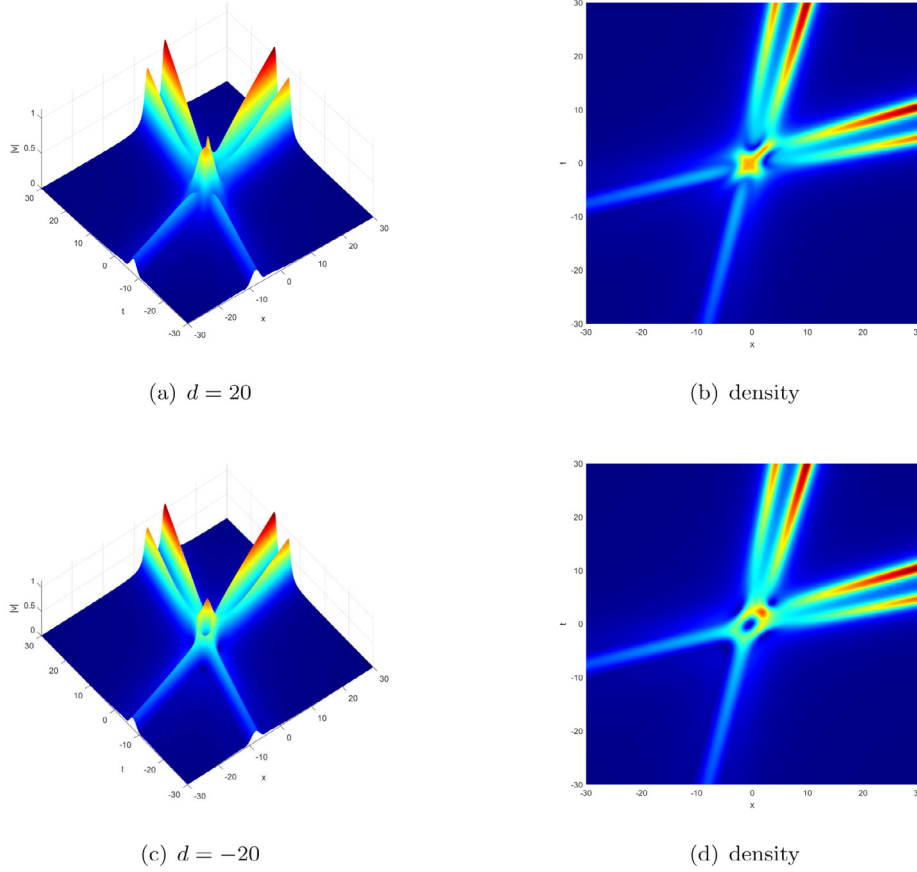


Fig. 18. High-order rational solution v and the corresponding density plot with parameters $t = 0$, $c = 20$, $\bar{p}_1 = -\frac{1}{2}$, $\bar{q}_1 = -2$, $a = 1$.

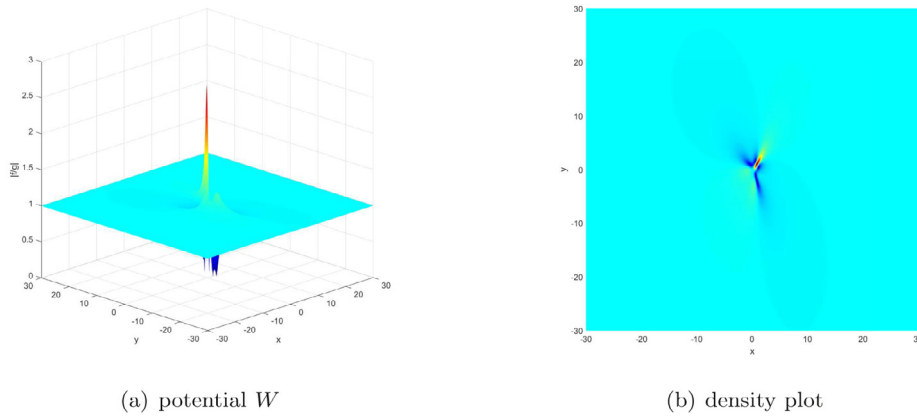


Fig. 19. High-order rational solution of the intermediate potential W at $t = 0$ with parameters $c = d = 0$, $a = 1$, $\bar{p}_1 = \bar{q}_1^* = 1 + i$ and the corresponding density plot.

6. Conclusion and discussion

The KP hierarchy reduction technique is a powerful method to investigate structures of solutions of integrable systems. Based on this approach, we investigated a (2+1)-dimensional generalized dispersive wave system. General soliton and (semi-)rational solutions are given in terms of $N \times N$ determinant. Under different size of the determinant solution, we construct soliton solutions, line breathers, algebraic solitons, multiple and high-order rational solutions on constant and periodic background, respectively. Dynamics of obtained solutions with different parameters are analyzed. Roles of constant a in DLW are discussed. As a byproduct, we find lump solutions for the intermediate potential of DLW system. It is known that the NLS equation possess multi-breather solutions on the elliptic function background. One question is how to construct breather solutions on the elliptic function background for DLW system. Meanwhile, nonlocal integrable equations, such as nonlocal NLS and nonlocal mKdV, were proposed from the reduction of well known AKNS system. These nonlocal systems possess $\mathcal{PT}$ symmetries and singular or nonsingular soliton solutions. High dimensional nonlocal integrable equations, for example, (2+1)-dimensional Davey–Stewartson equation, were also proposed. It would be interesting to investigate nonlocal DLW system and it remains a topic to be explored in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

When we investigate the collision of two algebraic solitons for DLW system in the case $|\xi| = O(|y|)$, we have the expression of v as follows

$$\begin{aligned}
 v &= 2 \left(\frac{f_{xy}}{f} - \frac{f_x f_y}{f^2} + \frac{g_{xy}}{g} - \frac{g_x g_y}{g^2} \right), \\
 \frac{f_{xy}}{f} &= - \frac{p^2 + q^2}{pq} \frac{1}{(1 + p\xi + c_{11} - \frac{p}{p+q})(-1 + q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2}} \\
 &= - \frac{p^2 + q^2}{pq} \frac{\xi^{-1}}{(p + (1 + c_{11} - \frac{p}{p+q})\xi^{-1})(-1 + q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \xi^{-1}} \\
 - \frac{f_x f_y}{f^2} &= - \frac{\left[q(1 + p\xi + c_{11} - \frac{p}{p+q}) + p(-1 + q\eta + d_{11} - \frac{q}{p+q}) \right] \left[-\frac{1}{q}(1 + p\xi + c_{11} - \frac{p}{p+q}) - \frac{1}{p}(-1 + q\eta + d_{11} - \frac{q}{p+q}) \right]}{\left[(1 + p\xi + c_{11} - \frac{p}{p+q})(-1 + q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \right]^2} \\
 &= - \frac{\left[q(p + (1 + c_{11} - \frac{p}{p+q})\xi^{-1}) + p(q\eta - 1 + d_{11} - \frac{q}{p+q})\xi^{-1} \right] \left[-\frac{(p + (1 + c_{11} - \frac{p}{p+q})\xi^{-1})}{q} - \frac{(-1 + q\eta + d_{11} - \frac{q}{p+q})}{p} \xi^{-1} \right]}{\left[(p + (1 + c_{11} - \frac{p}{p+q})\xi^{-1})(-1 + q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \xi^{-1} \right]^2} \\
 \frac{g_{xy}}{g} &= - \frac{p^2 + q^2}{pq} \frac{1}{(p\xi + c_{11} - \frac{p}{p+q})(q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2}} \\
 &= - \frac{p^2 + q^2}{pq} \frac{\xi^{-1}}{(p + (c_{11} - \frac{p}{p+q})\xi^{-1})(q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \xi^{-1}} \\
 - \frac{g_x g_y}{g^2} &= - \frac{\left[q(p\xi + c_{11} - \frac{p}{p+q}) + p(q\eta + d_{11} - \frac{q}{p+q}) \right] \left[-\frac{1}{q}(p\xi + c_{11} - \frac{p}{p+q}) - \frac{1}{p}(q\eta + d_{11} - \frac{q}{p+q}) \right]}{\left[(p\xi + c_{11} - \frac{p}{p+q})(q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \right]^2} \\
 &= - \frac{\left[q(p + (c_{11} - \frac{p}{p+q})\xi^{-1}) + p(q\eta + d_{11} - \frac{q}{p+q})\xi^{-1} \right] \left[-\frac{(p + (c_{11} - \frac{p}{p+q})\xi^{-1})}{q} - \frac{(q\eta + d_{11} - \frac{q}{p+q})}{p} \xi^{-1} \right]}{\left[(p + (c_{11} - \frac{p}{p+q})\xi^{-1})(q\eta + d_{11} - \frac{q}{p+q}) + \frac{pq}{(p+q)^2} \xi^{-1} \right]^2}.
 \end{aligned}$$

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